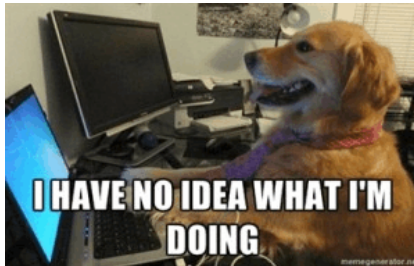


Por fin tenemos reference.



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1 Descripción de Algoritmos

1.1 Data structures

Segment tree — Complexity: $O(N_{build}, \log N_{peroperation})$

Usage: 1. `STree(int n)` → construye el arbol para n posiciones. 2. `upd(int p, int v)` → actualiza la posicion p con el valor v. 3. `query(int l, int r)` → devuelve el minimo en el rango [l, r). *Note:* Indexacion base 0; adaptar operacion neutra y combine para otro monoid.

Segment tree lazy — Complexity: $O(N_{build}, \log N_{perupdate/query})$

Usage: 1. `STree(int n)` → crea un arbol con lazy propagation. 2. `upd(int l, int r, int v)` → suma v al rango [l, r). 3. `query(int l, int r)` → devuelve la suma del rango [l, r). *Note:* Ejemplo de suma con incremento por rango; cambiar tipo de lazy segun el problema.

Persistent segment tree — Complexity: $O(\log N_{perversionupdate/query})$

Usage: 1. `STree(int n)` → reserva el arbol persistente sobre n posiciones. 2. `upd(int root, int p, int v)` → crea una nueva version cambiando la posicion p. 3. `query(int root, int l, int r)` → consulta el minimo en [l, r) para una version. *Note:* Mantener los indices de raiz de cada version por fuera.

2D segment tree — Complexity: $O(NM_{build}, \log N \log M_{peroperation})$

Usage: 1. `build()` → construye el arbol sobre la matriz global. 2. `upd(int x, int y, int v)` → actualiza una celda. 3. `query(int x0, int x1, int y0, int y1)` → consulta el rectangulo [x0, x1) x [y0, y1). *Note:* Usa arreglos globales; ajustar N y M antes de copiar.

Sparse table — Complexity: $O(N \log N_{build}, 1_{query})$

Usage: 1. `st_init(int *a)` → precalcula la tabla para el arreglo a. 2. `st_query(int s, int e)` → devuelve el minimo en [s, e). *Note:* Solo sirve para operaciones idempotentes como min/max/gcd.

Fenwick tree — Complexity: $O(\log N_{perupdate/query})$

Usage: 1. `upd(int i0, int v)` → suma v a la posicion i0. 2. `get(int i0)` → devuelve la suma de [0, i0). 3. `get_sum(int i0, int i1)` → devuelve la suma de [i0, i1). *Note:* API externa base 0; internamente usa base 1.

Wavelet tree — Complexity: $O(N \log \sigma_{build}, \log \sigma_{query})$

Usage: 1. `WT(int n)` → inicializa la estructura para n elementos. 2. `add(int v)` → agrega un valor comprimido al final. 3. `range(int k, int s, int e, int a, int b)` → cuenta valores en [a, b) dentro de posiciones [s, e). *Note:* Comprimir valores antes de insertar.

STL extended set — Complexity: $O(\log N_{peroperation})$

Usage: 1. `ordered_set` → set ordenado con orden estadístico. 2. `find_by_order(k)` → devuelve iterador al k-esimo elemento. 3. `order_of_key(x)` → cuenta elementos estrictamente menores que x. *Note:* Requiere GNU PBDS; no es C++ estandar puro.

STL rope — Complexity: $O(\log N_{persplit/insert/erase})$

Usage: 1. `rope<T>` → secuencia balanceada con operaciones de edición. 2. `insert(pos, value)` → inserta en una posición. 3. `erase(pos, len)` → borra un rango contiguo. *Note:* Requiere extensión GNU `__gnu_cxx::rope`.

Treap as BST — Complexity: $O(\log N_{expectedperoperation})$

Usage: 1. `split(pitem t, int key, pitem& l, pitem& r)` → separa por clave: $l \mid key, r \mid key$. 2. `insert(pitem& t, pitem it)` → inserta un nodo. 3. `kth(pitem t, int k)` → devuelve el nodo k -ésimo en orden. *Note:* Aleatorizado; crear nodos con prioridad random.

Implicit treap — Complexity: $O(\log N_{expectedperoperation})$

Usage: 1. `merge(pitem& t, pitem l, pitem r)` → concatena dos treaps. 2. `split(pitem t, pitem& l, pitem& r, int sz)` → deja sz elementos en l . 3. `output(pitem t)` → recorre la secuencia en orden. *Note:* Las claves son posiciones implícitas; propagar lazy con push.

Implicit treap with father — Complexity: $O(\log N_{expectedperoperation})$

Usage: 1. `merge(pitem& t, pitem l, pitem r)` → concatena y actualiza padres. 2. `split(pitem t, pitem& l, pitem& r, int sz)` → separa por tamaño. 3. `root(pitem t, int& pos)` → devuelve raíz y posición actual del nodo. *Note:* Util cuando se conservan punteros a nodos individuales.

Link-cut tree — Complexity: $O(\log N_{amortizedperoperation})$

Usage: 1. `evert(pitem v)` → convierte v en raíz del árbol representado. 2. `link(pitem v, pitem w)` → agrega arista haciendo v hijo de w . 3. `cut(pitem v, pitem w)` → elimina la arista entre v y w . *Note:* Antes de usar, inicializar campos de cada nodo.

Link-cut tree with path aggregate — Complexity:

$O(\log N_{amortizedperoperation})$

Usage: 1. `mkR(Node x)` → hace a x raíz representada. 2. `link(Node x, Node y)` → conecta dos componentes. 3. `query(Node x, Node y)` → consulta el agregado del camino x - y . *Note:* Modificar `mOp`, `qOp` y deltas para el agregado requerido.

Static convex hull trick — Complexity: $O(\log N_{perqueryaftermonotoneinsert})$

Usage: 1. `CHT::add(Line l)` → agrega una recta al hull. 2. `CHT::eval(int p, tc x)` → evalúa la recta p en x . 3. `CHT::query(tc x)` → devuelve el mínimo en x . *Note:* Para máximo, cambiar signos como indica el comentario del archivo.

Dynamic convex hull trick — Complexity: $O(\log N_{perinsert/query})$

Usage: 1. `HullDynamic::add_line(tc m, tc b)` → agrega $y = mx + b$. 2. `HullDynamic::eval(Line l, tc x)` → evalúa una recta. 3. `HullDynamic::query(tc x)` → devuelve el máximo en x . *Note:* Implementado para máximo.

Li Chao tree — Complexity: $O(\log C_{perinsert/query})$

Usage: 1. `LiChaoTree(T xmin, T xmax)` → crea el dominio discreto/entero $[xmin, xmax]$. 2. `add_line(Line nw)` → inserta una recta. 3. `query(T x)` → devuelve el mínimo en x . *Note:* Ajustar INF y dominio al rango de coordenadas.

Gain-cost set — Complexity: $O(\log N_{amortizedperinsert/query})$

Usage: 1. `GCS::add(ii x)` → agrega un par y elimina dominados. 2. `GCS::query(int v)` → calcula el mejor valor para v según los pares activos. *Note:* Mantiene frontera no dominada en un set.

Disjoint intervals — Complexity: $O(K \log N_{perinsertion})$

Usage: 1. `disjoint_intervals::add(ii v)` → agrega el intervalo y une solapados. 2. `set<ii> s` → contiene intervalos disjuntos ordenados. *Note:* Intervalos semiarbitrarios según ii ; revisar inclusividad antes de copiar.

1.2 Graphs

Topological sort — Complexity: $O(V + E)$

Usage: 1. `tsort()` → devuelve un orden topológico lexicográficamente mínimo. 2. `priority_queue<int, vector<int>, greater<int>>` → elige siempre el nodo libre más chico. *Note:* Si el resultado tiene menos de n nodos, hay ciclo.

Kruskal MST — Complexity: $O(E \log E)$

Usage: 1. `uf_init()` → inicializa union-find. 2. `uf_join(int x, int y)` → une componentes si eran distintas. 3. `kruskal()` → devuelve el peso del MST asumiendo grafo conexo. *Note:* Ordenar aristas por peso antes de procesarlas si el archivo no lo hace por ti.

Dijkstra — Complexity: $O((V + E) \log V)$

Usage: 1. `dijkstra(int x)` → calcula distancias mínimas desde x . 2. `dist[v]` → queda con -1 si v no es alcanzable. *Note:* Requiere pesos no negativos.

Bellman-Ford — Complexity: $O(VE)$

Usage: 1. `bford(int src)` → calcula distancias desde `src` y propaga ciclos negativos. 2. `d[v]` → guarda INF, distancia finita o -INF según alcanzabilidad. *Note:* Sirve con pesos negativos; detecta nodos afectados por ciclos negativos alcanzables.

Floyd-Warshall — Complexity: $O(V^3)$

Usage: 1. `floyd()` → reemplaza g por distancias mínimas entre todos los pares. 2. `inNegCycle(int v)` → indica si v está en un ciclo negativo. 3. `hasNegCycle(int a, int b)` → indica si hay ciclo negativo en algún camino a - b . *Note:* Inicializar $g[i][i]=0$ y $g[u][v]=\text{peso}$.

SCC and 2-SAT — Complexity: $O(V + E)$

Usage: 1. `scc()` → calcula componentes fuertemente conexas. 2. `addor(int a, int b)` → agrega la cláusula $a \text{ OR } b$. 3. `satisf(int nvar)` → devuelve si la instancia 2-SAT es satisfacible. *Note:* Usa literales codificados con `neg(a)`.

Articulation, bridges and biconnected — Complexity: $O(V + E)$

Usage: 1. `add_edge(int u, int v)` → agrega una arista no dirigida. 2. `doit()` → ejecuta DFS y marca puentes/componentes. 3. `edge.bridge` → queda verdadero para aristas puente. *Note:* Reinicializar vectores globales entre casos.

Chu-Liu Edmonds arborescence — Complexity: $O(VE)$

Usage: 1. `ChuLiu(int n, int r)` → crea instancia con n nodos y raíz r. 2. `add_edge(int u, int v, tw w)` → agrega arco dirigido u → v. 3. `doit()` → devuelve el costo de la arborescencia mínima. *Note:* Los nodos deben ser alcanzables desde la raíz.

LCA binary lifting — Complexity: $O(N \log N_{build}, \log N_{query})$

Usage: 1. `lca_init()` → precalcula profundidad y tabla binaria. 2. `lca(int x, int y)` → devuelve el ancestro comun mas bajo. *Note:* Asume arbol enraizado en 0.

Heavy-light decomposition — Complexity: $O(N_{build}, \log^2 N_{query})$

Usage: 1. `hld_init()` → calcula cadenas y posiciones. 2. `query(int x, int y, STree& rmq)` → consulta el camino x-y usando segment tree. *Note:* Combinar con un segment tree sobre las posiciones linealizadas.

Centroid decomposition — Complexity: $O(N \log N)$

Usage: 1. `centroid()` → construye la descomposicion. 2. `cdfs(int x, int f, int sz)` → procesa recursivamente centroides. *Note:* Completar la logica de actualizacion/consulta dentro del DFS segun el problema.

Parallel DFS — Complexity: $O(V + E)$

Usage: 1. `Tree` → estructura auxiliar para DFS en paralelo. 2. `Tree::go(int x)` → recorre el arbol manteniendo estado compartido. *Note:* Snippet especializado; revisar invariantes del problema antes de usar.

Eulerian path — Complexity: $O(V + E)$

Usage: 1. `add_edge(int a, int b)` → agrega arista al multigrafo. 2. `get_path(int x)` → devuelve un camino euleriano que empieza en x. *Note:* Verificar grados/conectividad antes de llamar si el problema no lo garantiza.

Offline dynamic connectivity — Complexity: $O((N + Q) \log Q)$

Usage: 1. `DynCon(int n, int q)` → prepara estructura offline. 2. `add(int l, int r, Query e)` → activa una arista en el intervalo [l, r]. 3. `go(int k, int s, int e)` → resuelve recursivamente consultas con rollback. *Note:* Procesa consultas offline con union-find rollback.

Dominator tree — Complexity: $O((V + E)\alpha(V))$

Usage: 1. `dominators(int rt)` → calcula dominadores desde la raíz rt. 2. `idom[v]` → padre inmediato de v en el arbol de dominadores. *Note:* Algoritmo Lengauer-Tarjan; nodos no alcanzables requieren filtrado.

Edmonds blossom — Complexity: $O(V^2E)$

Usage: 1. `edmonds()` → devuelve tamaño de matching máximo en grafo general. 2.

`aug(int s, int t)` → busca camino aumentante desde s. *Note:* Grafo no dirigido; limpiar matching global entre casos.

1.3 Math

Integer floor division — Complexity: $O(1)$

Usage: 1. `floordiv(ll x, ll y, ll& q, ll& r)` → calcula $q = \text{floor}(x/y)$ y residuo r. 2. `q, r` → satisfacen $x = q*y + r$ con r ajustado al signo esperado. *Note:* Pensado para x negativo; revisar convencion de residuo si $y \neq 0$.

Extended Euclid — Complexity: $O(\log \min(a, b))$

Usage: 1. `euclid(ll a, ll b, ll& x, ll& y)` → devuelve gcd(a,b) y coeficientes x,y. 2. `x, y` → cumplen $a*x + b*y = \text{gcd}(a,b)$.

Sieve of Eratosthenes — Complexity: $O(N \log \log N)$

Usage: 1. `init_sieve()` → precalcula primalidad/factor menor hasta MAXN. 2. `sv[x]` → indica factor o estado de x segun el snippet. *Note:* Ajustar MAXN antes de compilar.

Generate divisors — Complexity: $O(d(n))$

Usage: 1. `divisors(vector<pair<ll,int>> f)` → genera todos los divisores desde la factorizacion. 2. `div_rec(...)` → recursion interna para multiplicar potencias. *Note:* La entrada es factorizacion primo, exponente.

Pollard Rho factorization — Complexity: $O(\text{probabilistic}, \text{about } n^{1/4})$

Usage: 1. `rabin(ll n)` → test probabilistico/deterministico practico de primalidad. 2. `rho(ll n)` → encuentra un factor no trivial. 3. `fact(ll n, map<ll,int>& f)` → factoriza n en el mapa f. *Note:* Para la version con primos chicos, llamar `init_sv` primero.

Simpson integration — Complexity: $O(N)$

Usage: 1. `integrate(double f(double), double a, double b, int n)` → aproxima la integral de f en [a,b]. 2. `n` → cantidad de subintervalos; mayor n mejora precision. *Note:* Usar n par/grande para mejor estabilidad.

Polynomial utilities — Complexity: $O(\text{dependsonoperation})$

Usage: 1. `poly<T>` → representa polinomios por coeficientes. 2. `roots(poly<> p)` → busca raíces enteras. 3. `interpolate(vector<tp> x, vector<tp> y)` → construye polinomio interpolante. *Note:* Complejidades son cuadraticas en las operaciones basicas incluidas.

Bairstow roots — Complexity: $O(ITER * \text{degree})$

Usage: 1. `bairstow(poly<> p)` → aproxima un factor cuadrático de p. 2. `roots(poly<> p)` → devuelve raíces reales aproximadas. *Note:* Requiere el tipo `poly` del archivo de polinomios.

Fast Fourier Transform — Complexity: $O(N \log N)$

Usage: 1. `dft(CD* a, int n, bool inv)` → transforma in-place. 2.

`multiply(poly& p1, poly& p2)` → multiplica polinomios/convolucionadores.
Note: Elegir version FFT o NTT segun comentarios del archivo.

FFT polynomial operations — Complexity:

$O(N \log N \text{ per high-level operation})$

Usage: 1. `invert(poly& p, int d)` → inverso formal modulo x^d . 2. `log(poly& p, int d)` → logaritmo formal modulo x^d . 3. `interpolate(vector<tf>& x, vector<tf>& y)` → interpola puntos usando arbol de productos. *Note:* Depende de `multiply` y aritmetica modular definida en el notebook.

Fast Hadamard Transform — Complexity: $O(N \log N)$

Usage: 1. `fht(ll* p, int n, bool inv)` → transforma para XOR/AND/OR segun variante. 2. `multiply(vector<ll>& p1, vector<ll>& p2)` → convolucion con la transformada elegida. *Note:* Ajustar el operador dentro del archivo segun XOR, AND u OR.

Karatsuba multiplication — Complexity: $O(N^{1.585})$

Usage: 1. `karatsura(int n, tp* p, tp* q, tp* r)` → multiplica arreglos de tamano n. 2. `multiply(vector<tp> p0, vector<tp> p1)` → devuelve la convolucion. *Note:* El nombre de la funcion interna esta escrito como en el snippet.

Linear Diophantine equations — Complexity: $O(\log \min(a, b))$

Usage: 1. `extendedEuclid(ll a, ll b)` → devuelve coeficientes de Bezout. 2. `diophantine(ll a, ll b, ll r)` → resuelve $a*x + b*y = r$ y direccion de soluciones. *Note:* r debe ser multiplo de $\gcd(a, b)$.

Modular inverse — Complexity: $O(\log \text{mod})$

Usage: 1. `inv(ll a, ll mod)` → devuelve el inverso de a modulo mod. *Note:* Requiere $\gcd(a, \text{mod})=1$.

Chinese remainder theorem — Complexity: $O(K \log M)$

Usage: 1. `sol(tuple<ll, ll, ll> c)` → resuelve una congruencia $a*x = b \text{ mod } m$. 2. `crt(vector<tuple<ll, ll, ll>> cond)` → combina condiciones y devuelve solucion, lcm. *Note:* Requiere `inv`, `diophantine`, `gcd` y que el lcm quepa en ll.

Mobius function — Complexity: $O(N \log N)$

Usage: 1. `mobius()` → precalcula mu hasta MAXN. 2. `mu[i]` → queda con la funcion de Mobius de i. *Note:* Ajustar MAXN y limpiar arreglos entre usos.

Matrix fast exponentiation — Complexity: $O(N^3 \log E)$

Usage: 1. `ones(int n)` → devuelve matriz identidad n x n. 2. `be(Matrix b, ll e)` → calcula b^e por exponenciacion binaria. *Note:* Cambiar multiplicacion/modulo dentro del operador si hace falta.

Gaussian elimination — Complexity: $O(NM \min(N, M))$

Usage: 1. `reduce(vector<vector<double>>& x)` → reduce la matriz y devuelve determinante cuando aplica. 2. `x` → queda en forma escalonada reducida. *Note:* Usa `double` y EPS; para modulares cambiar division y comparacion.

Simplex — Complexity: $O(\text{exponential worst-case, fast in practice})$

Usage: 1. `simplex(int n, int m)` → resuelve LP en forma estandar. 2. `pivot(int x, int y)` → realiza pivoteo interno. *Note:* Revisar convencion de restricciones antes de copiar.

Discrete logarithm — Complexity: $O(\sqrt{m})$

Usage: 1. `discrete_log(ll a, ll b, ll m)` → encuentra x tal que $a^x \equiv b \pmod{m}$. *Note:* Baby-step giant-step; revisar casos $\gcd(a, m) \neq 1$.

Berlekamp-Massey — Complexity: $O(N^2)$

Usage: 1. `BM(vi x)` → devuelve coeficientes de la recurrencia minima. 2. `cur` → coeficientes normalizados modulo MOD. *Note:* La secuencia y MOD son globales/externos.

Linear recurrence — Complexity: $O(K^2 \log N)$

Usage: 1. `LinearRec(vector<int> terms, vector<int> trans)` → define terminos iniciales y transicion. 2. `calc(ll k)` → devuelve el k-esimo termino. *Note:* Combina Berlekamp-Massey con exponenciacion de polinomios.

Tonelli-Shanks — Complexity: $O(\log^2 p)$

Usage: 1. `legendre(ll a, ll p)` → calcula simbolo de Legendre. 2. `tonelli_shanks(ll n, ll p)` → devuelve una raiz cuadrada de n modulo p. *Note:* p debe ser primo.

Prime counting — Complexity: $O(n^{2/3} / \log n)$

Usage: 1. `prime_count(ll n)` → devuelve $\pi(n)$, cantidad de primos $j \leq n$. *Note:* Usar ll; revisar memoria para n grande.

Number Theoretic Transform — Complexity: $O(N \log N)$

Usage: 1. `FFT<u, uu, p, root>` → transformada modular parametrizada. 2. `conv_small(const poly& a, const poly& b)` → convolucion bajo modulo NTT. 3. `conv_sunzi(const poly& a, const poly& b, ll m)` → convolucion con CRT para modulo arbitrario. *Note:* Elegir primos/raices compatibles con el tamano.

Lattice points under line — Complexity: $O(\log \max(a, b, c))$

Usage: 1. `f(ll a, ll b, ll c)` → suma pisos relacionada con $ax+b$ sobre c. 2. `g(ll a, ll b, ll c, ll X, ll Y)` → cuenta puntos bajo una recta en una caja. *Note:* Verificar inclusividad de bordes para el problema.

1.4 Geometry

Point — Complexity: $O(1 \text{ per operation})$

Usage: 1. `pt` → punto/vector 2D con operaciones aritmeticas. 2. `pt::norm()` → devuelve longitud. 3. `pt::rot(pt r)` → rota usando el vector r. *Note:* Para 3D, agregar coordenada z como indica el comentario.

Line — Complexity: $O(1 \text{ per operation})$

Usage: 1. `ln(pt p, pt q)` → crea recta por dos puntos. 2. `ln::proj(pt p)` →

proyecta p sobre la recta. 3. `operator^(ln l, ln m)` → interseccion de dos rectas. *Note:* Depende de pt y EPS.

Circle — Complexity: $O(1 \text{ per primitive}, N \text{ for intercircles})$

Usage: 1. `circle(pt o, double r)` → define centro y radio. 2. `circle::has(pt p)` → clasifica punto respecto al circulo. 3. `intercircles(vector<circle> c)` → calcula area cubierta por circulos. *Note:* Usa EPS; revisar tangencias.

Polygon — Complexity: $O(N \text{ per basic query}, \log^2 N \text{ dynamichull query})$

Usage: 1. `pol::area()` → devuelve area orientada. 2. `pol::has(pt q)` → clasifica punto respecto del poligono. 3. `add(pt q) / query(pt v)` → mantiene hull dinamico y consulta maximo dot. *Note:* Convenciones de orientacion y borde dependen de sgn/EPS.

Plane — Complexity: $O(1 \text{ per operation})$

Usage: 1. `plane(pt p, pt n)` → plano por punto y normal. 2. `plane::side(pt q)` → clasifica q respecto al plano. 3. `operator^(plane a, plane b)` → interseccion de planos cuando aplica. *Note:* Pensado para geometria 3D con pt extendido.

Radial order — Complexity: $O(N \log N \text{ sort})$

Usage: 1. `Cmp` → comparador angular para ordenar puntos radialmente. 2. `sort(v.begin(), v.end(), Cmp())` → ordena por semiplano y producto cruz. *Note:* El comentario del archivo pide const en operator- de pt.

Convex hull — Complexity: $O(N \log N)$

Usage: 1. `chull(vector<pt> p)` → devuelve el hull convexo de los puntos. *Note:* Revisar si conserva o elimina colineales segun la condicion de giro.

Planar graph dual — Complexity: $O(E \log E)$

Usage: 1. `get_dual(vector<pt> p)` → construye caras/dual desde embedding planar. 2. `g[u]` → lista de adyacencia geometrica del grafo primal. *Note:* Los puntos deben corresponder a nodos del embedding sin cruces.

Halfplane intersection — Complexity: $O(N \log N)$

Usage: 1. `halfplane(pt p, pt q)` → define semiplano por borde orientado. 2. `intersect(vector<halfplane> b)` → devuelve poligono de interseccion. *Note:* Agregar caja grande si la interseccion puede ser no acotada.

KD tree — Complexity: $O(N \log N \text{ build}, \text{about } \log N \text{ query average})$

Usage: 1. `KDTree(vector<pt> p)` → construye KD-tree. 2. `nearest(pt q)` → consulta punto mas cercano. *Note:* Rendimiento depende de distribucion de puntos.

Minkowski sum — Complexity: $O(N + M \text{ after sorting})$

Usage: 1. `reorder(vector<pt>& p)` → rota poligono para empezar por el menor punto. 2. `minkowski_sum(vector<pt> p, vector<pt> q)` → devuelve suma de Minkowski de poligonos convexos. *Note:* Los poligonos deben estar en orden alrededor del borde.

Delaunay triangulation — Complexity: $O(N \log N \text{ expected})$

Usage: 1. `delaunay(vector<pt> v)` → devuelve triangulacion como adyacencias. 2. `in_c(pt a, pt b, pt c, pt p)` → test de circuncirculo interno. *Note:* Evitar puntos duplicados; sensible a EPS/orientacion.

1.5 Strings

KMP — Complexity: $O(N + M)$

Usage: 1. `kmppre(string& t)` → calcula funcion de prefijo/bordes del patron. 2. `kmp(string& s, string& t)` → encuentra ocurrencias de t en s. *Note:* Adaptar el cuerpo de kmp para guardar/imprimir matches.

Z function — Complexity: $O(N)$

Usage: 1. `z_function(string& s)` → devuelve z[i], largo del prefijo que empieza en i. *Note:* Util para matching concatenando patron + separador + texto.

Manacher — Complexity: $O(N)$

Usage: 1. `manacher(string& s)` → precalcula radios de palindromos pares e impares. 2. `d1, d2` → arreglos de radios segun paridad. *Note:* Revisar convencion de indices del snippet.

Aho-Corasick — Complexity: $O(\text{total length} + \text{matches})$

Usage: 1. `aho_init()` → inicializa el automata. 2. `add_string(string s, int id)` → agrega patron con identificador. 3. `go(int v, char c)` → transicion del automata con links calculados lazy. *Note:* No olvidar llamar aho.init antes de insertar.

Suffix automaton — Complexity: $O(N)$

Usage: 1. `sa_init()` → inicializa automata vacio. 2. `sa_extend(char c)` → agrega un caracter al final. 3. `state::next` → transiciones por caracter. *Note:* Limpiar next en cada estado nuevo.

Palindromic tree — Complexity: $O(N)$

Usage: 1. `palindromic_tree` → estructura Eertree para palindromos distintos. 2. `add(int c)` → agrega un caracter y actualiza sufijo palindromico. 3. `node` → almacena longitud, link y transiciones. *Note:* Usar alfabeto/offset adecuados.

Suffix array slow — Complexity: $O(N \log^2 N)$

Usage: 1. `constructSA(string& s)` → devuelve suffix array usando comparacion de substrings. 2. `sacomp(int lhs, int rhs)` → comparador de sufijos. *Note:* Simple pero lento; preferir la version $O(N \log N)$ para strings grandes.

Suffix array — Complexity: $O(N \log N)$

Usage: 1. `constructSA(string& s)` → construye suffix array. 2. `csort(vector<int>& sa, vector<int>& r, int k)` → counting sort por ranking desplazado. *Note:* Agregar sentinel si el problema lo requiere.

LCP array — Complexity: $O(N)$

Usage: 1. `computeLCP(string& s, vector<int>& sa)` → devuelve LCP entre sufijos consecutivos en sa. *Note:* Requiere suffix array valido de s.

Suffix tree — Complexity: $O(N)$

Usage: 1. `SuffixTree(string s)` → construye arbol de sufijos con Ukkonen. 2. `go(...)` → navega por transiciones internas. 3. `split(...)` → divide una arista durante la extension. *Note:* Agregar sentinel unico al string para hojas explicitas.

String hashing — Complexity: $O(N_{build}, 1_{query})$

Usage: 1. `Hash(string s)` → precalcula hashes/potencias. 2. `get(int l, int r)` → devuelve hash del substring $[l, r)$. *Note:* Usar doble hash o modulo robusto si hay riesgo de colision.

String hashing with `_int128` — Complexity: $O(N_{build}, 1_{query})$

Usage: 1. `Hash(string s)` → precalcula hash con multiplicacion amplia. 2. `get(int l, int r)` → devuelve hash de substring. *Note:* Usa `_int128`; requiere compilador GNU/Clang compatible.

1.6 Flow**Bipartite matching DFS — Complexity: $O(VE)$**

Usage: 1. `match(int x)` → intenta aumentar desde el nodo izquierdo x. 2. `mm()` → devuelve las parejas encontradas. *Note:* Mas simple que Hopcroft-Karp; usar en grafos chicos.

Hopcroft-Karp — Complexity: $O(E\sqrt{V})$

Usage: 1. `bfs()` → construye capas desde libres izquierdos. 2. `dfs(int x)` → aumenta por caminos de capas. 3. `mm()` → devuelve el tamaño del matching maximo. *Note:* Particiones izquierda/derecha definidas por los arreglos globales.

Hungarian algorithm — Complexity: $O(N^3)$

Usage: 1. `Hungarian(int n, int m)` → resuelve asignacion rectangular. 2. `assign()` → devuelve costo y matching optimo. 3. `L[i]` → columna asignada a la fila i. *Note:* Cambiar signo de costos para maximizacion.

Dinic max flow — Complexity: $O(V^2E)$

Usage: 1. `Dinic(int nodes)` → crea red con nodes vertices. 2. `add_edge(int s, int t, ll cap)` → agrega arco con capacidad. 3. `max_flow(int src, int dst)` → devuelve flujo maximo src-dst. *Note:* Despues del flujo, dist permite recuperar corte minimo como indica el comentario.

Min-cost max-flow — Complexity: $O(FE \log V)$

Usage: 1. `MCF(int n)` → crea red con n vertices. 2. `add_edge(int s, int t, tf cap, tc cost)` → agrega arco capacitado con costo. 3. `get_flow(int s, int t)` → devuelve {flujo, costo} maximo de costo minimo. *Note:* Usa potenciales y Dijkstra; costos negativos requieren potencial inicial valido.

1.7 Other**Mo algorithm — Complexity: $O((N + Q)\sqrt{N})$**

Usage: 1. `mos()` → ordena consultas y mantiene ventana activa. 2. `add(pos) / remove(pos)` → actualizan la respuesta al mover extremos. 3. `qu{l,r,id}` → representa consulta offline. *Note:* Completar add/remove segun la funcion objetivo.

Divide and conquer DP optimization — Complexity: $O(KN \log N)$

Usage: 1. `doit(int s, int e, int s0, int e0, int i)` → calcula una capa DP restringiendo optimo. 2. `doall()` → ejecuta todas las capas necesarias. *Note:* Valido si el optimo es monotono.

Date conversion — Complexity: $O(1)$

Usage: 1. `dateToInt(int y, int m, int d)` → convierte fecha gregoriana a entero juliano. 2. `intToDate(int jd, int& y, int& m, int& d)` → convierte entero a fecha. 3. `DayOfWeek(int d, int m, int y)` → devuelve dia de semana empezando en domingo. *Note:* Revisar calendario esperado por el problema.

2 Data structures

2.1 Segment tree

```

1 #define oper min
2 #define NEUT INF
3 struct STree { // segment tree for min over integers
4     vector<int> st;int n;
5     STree(int n): st(4*n+5,NEUT), n(n) {}
6     void init(int k, int s, int e, int *a){
7         if(s+1==e){st[k]=a[s];return;}
8         int m=(s+e)/2;
9         init(2*k,s,m,a);init(2*k+1,m,e,a);
10        st[k]=oper(st[2*k],st[2*k+1]);
11    }
12    void upd(int k, int s, int e, int p, int v){
13        if(s+1==e){st[k]=v;return;}
14        int m=(s+e)/2;
15        if(p<m)upd(2*k,s,m,p,v);
16        else upd(2*k+1,m,e,p,v);
17        st[k]=oper(st[2*k],st[2*k+1]);
18    }
19    int query(int k, int s, int e, int a, int b){
20        if(s>=b||e<=a)return NEUT;
21        if(s>=a&&e<=b)return st[k];
22        int m=(s+e)/2;
23        return oper(query(2*k,s,m,a,b),query(2*k+1,m,e,a,b));
24    }
25    void init(int *a){init(1,0,n,a);}
26    void upd(int p, int v){upd(1,0,n,p,v);}
27    int query(int a, int b){return query(1,0,n,a,b);}
28 }; // usage: STree rmq(n);rmq.init(x);rmq.upd(i,v);rmq.query(s,e);

```

2.2 Segment tree - Lazy propagation

```

1 struct STree { // example: range sum with range addition
2     vector<int> st,lazy;int n;
3     STree(int n): st(4*n+5,0), lazy(4*n+5,0), n(n) {}
4     void init(int k, int s, int e, int *a){
5         lazy[k]=0; // lazy neutral element
6         if(s+1==e){st[k]=a[s];return;}
7         int m=(s+e)/2;
8         init(2*k,s,m,a);init(2*k+1,m,e,a);

```

```

9         st[k]=st[2*k]+st[2*k+1]; // operation
10    }
11    void push(int k, int s, int e){
12        if(!lazy[k])return; // if neutral, nothing to do
13        st[k]+=(e-s)*lazy[k]; // update st according to lazy
14        if(s+1<e){ // propagate to children
15            lazy[2*k]+=lazy[k];
16            lazy[2*k+1]+=lazy[k];
17        }
18        lazy[k]=0; // clear node lazy
19    }
20    void upd(int k, int s, int e, int a, int b, int v){
21        push(k,s,e);
22        if(s>=b||e<=a)return;
23        if(s>=a&&e<=b){
24            lazy[k]+=v; // accumulate lazy
25            push(k,s,e);return;
26        }
27        int m=(s+e)/2;
28        upd(2*k,s,m,a,b,v);upd(2*k+1,m,e,a,b,v);
29        st[k]=st[2*k]+st[2*k+1]; // operation
30    }
31    int query(int k, int s, int e, int a, int b){
32        if(s>=b||e<=a)return 0; // operation neutral
33        push(k,s,e);
34        if(s>=a&&e<=b)return st[k];
35        int m=(s+e)/2;
36        return query(2*k,s,m,a,b)+query(2*k+1,m,e,a,b); // operation
37    }
38    void init(int *a){init(1,0,n,a);}
39    void upd(int a, int b, int v){upd(1,0,n,a,b,v);}
40    int query(int a, int b){return query(1,0,n,a,b);}
41 }; // usage: STree rmq(n);rmq.init(x);rmq.upd(s,e,v);rmq.query(s,e);

```

2.3 Segment tree - Persistence

```

1 #define oper(a,b) min(a,b)
2 #define NEUT INF
3 struct STree { // persistent segment tree for min over integers
4     vector<int> st, L, R; int n,sz,rt;
5     STree(int n): st(1,NEUT),L(1,0),R(1,0),n(n),rt(0),sz(1){}
6     int new_node(int v, int l=0, int r=0){
7         int ks=sz(st);

```



```

8   st.push_back(v);L.push_back(l);R.push_back(r);
9   return ks;
10  }
11  int init(int s, int e, int *a){ // not necessary in most cases
12      if(s+1==e)return new_node(a[s]);
13      int m=(s+e)/2,l=init(s,m,a),r=init(m,e,a);
14      return new_node(oper(st[l],st[r]),l,r);
15  }
16  int upd(int k, int s, int e, int p, int v){
17      int ks=new_node(st[k],L[k],R[k]);
18      if(s+1==e){st[k]=v;return ks;}
19      int m=(s+e)/2,ps;
20      if(p<m)ps=upd(L[ks],s,m,p,v),L[ks]=ps;
21      else ps=upd(R[ks],m,e,p,v),R[ks]=ps;
22      st[ks]=oper(st[L[ks]],st[R[ks]]);
23      return ks;
24  }
25  int query(int k, int s, int e, int a, int b){
26      if(e<=a||b<=s)return NEUT;
27      if(a<=s&&e<=b)return st[k];
28      int m=(s+e)/2;
29      return oper(query(L[k],s,m,a,b),query(R[k],m,e,a,b));
30  }
31  int init(int *a){return init(0,n,a);}
32  int upd(int k, int p, int v){return rt=upd(k,0,n,p,v);}
33  int upd(int p, int v){return upd(rt,p,v);} // update on last root
34  int query(int k,int a, int b){return query(k,0,n,a,b);}
35  }; // usage: STree rmq(n);root=rmq.init(x);new_root=rmq.upd(root,i,v);
      rmq.query(root,s,e);

```

2.4 Segment tree - 2D

```

1  int n,m;
2  int a[MAXN][MAXN],st[2*MAXN][2*MAXN];
3  void build(){
4      for (int i = 0; i < n; ++i)for (int j = 0; j < m; ++j)st[i+n][j+m]=a[i][j];
5      for (int i = 0; i < n; ++i)for(int j=m-1;j--;j)
6          st[i+n][j]=op(st[i+n][j<<1],st[i+n][j<<1|1]);
7      for(int i=n-1;i--;i)for (int j = 0; j < 2*m; ++j)
8          st[i][j]=op(st[i<<1][j],st[i<<1|1][j]);
9  }
10 void upd(int x, int y, int v){

```

```

11  st[x+n][y+m]=v;
12  for(int j=y+m;j>1;j>>=1)st[x+n][j>>1]=op(st[x+n][j],st[x+n][j^1]);
13  for(int i=x+n;i>1;i>>=1)for(int j=y+m;j;j>>=1)
14      st[i>>1][j]=op(st[i][j],st[i^1][j]);
15  }
16  int query(int x0, int x1, int y0, int y1){
17      int r=NEUT;
18      for(int i0=x0+n,i1=x1+n;i0<i1;i0>>=1,i1>>=1){
19          int t[4],q=0;
20          if(i0&1)t[q++]=i0++;
21          if(i1&1)t[q++]--i1;
22          for (int k = 0; k < q; ++k)for(int j0=y0+m,j1=y1+m;j0<j1;j0>>=1,j1
23              >>=1){
24              if(j0&1)r=op(r,st[t[k]][j0++]);
25              if(j1&1)r=op(r,st[t[k]][--j1]);
26          }
27      }
28      return r;

```

2.5 Sparse table (static RMQ)

```

1  #define oper min
2  int st[K][1<<K];int n; // K such that 2^K>n
3  void st_init(int *a){
4      for (int i = 0; i < n; ++i)st[0][i]=a[i];
5      for (int k = 1; k < K; ++k)for (int i = 0; i < n-(1<<k)+1; ++i)
6          st[k][i]=oper(st[k-1][i],st[k-1][i+(1<<(k-1))]);
7  }
8  int st_query(int s, int e){
9      int k=31-__builtin_clz(e-s);
10     return oper(st[k][s],st[k][e-(1<<k)]);
11 }

```

2.6 Fenwick tree

```

1  int ft[MAXN+1]; // for more dimensions, make ft multi-dimensional
2  void upd(int i0, int v){ // add v to i0th element (0-based)
3      // add extra fors for more dimensions
4      for(int i=i0+1;i<=MAXN;i+=i&-i)ft[i]+=v;
5  }
6  int get(int i0){ // get sum of range [0,i0]
7      int r=0;
8      // add extra fors for more dimensions

```

```

9   for(int i=i0;i;i-=i&&-i)r+=ft[i];
10  return r;
11 }
12 int get_sum(int i0, int i1){ // get sum of range [i0,i1) (0-based)
13     return get(i1)-get(i0);
14 }

```

2.7 Wavelet tree

```

1  struct WT {
2      vector<int> wt[1<<20];int n;
3      void init(int k, int s, int e){
4          if(s+1==e)return;
5          wt[k].clear();wt[k].push_back(0);
6          int m=(s+e)/2;
7          init(2*k,s,m);init(2*k+1,m,e);
8      }
9      void add(int k, int s, int e, int v){
10         if(s+1==e)return;
11         int m=(s+e)/2;
12         if(v<m)wt[k].push_back(wt[k].back()),add(2*k,s,m,v);
13         else wt[k].push_back(wt[k].back()+1),add(2*k+1,m,e,v);
14     }
15     int query0(int k, int s, int e, int a, int b, int i){
16         if(s+1==e)return s;
17         int m=(s+e)/2;
18         int q=(b-a)-(wt[k][b]-wt[k][a]);
19         if(i<q)return query0(2*k,s,m,a-wt[k][a],b-wt[k][b],i);
20         else return query0(2*k+1,m,e,wt[k][a],wt[k][b],i-q);
21     }
22     void upd(int k, int s, int e, int i){
23         if(s+1==e)return;
24         int m=(s+e)/2;
25         int v0=wt[k][i+1]-wt[k][i],v1=wt[k][i+2]-wt[k][i+1];
26         if(!v0&&!v1)upd(2*k,s,m,i-wt[k][i]);
27         else if(v0&&v1)upd(2*k+1,m,e,wt[k][i]);
28         else if(v0)wt[k][i+1]--;
29         else wt[k][i+1]++;
30     }
31     void init(int _n){n=_n;init(1,0,n);} // (values in range [0,n))
32     void add(int v){add(1,0,n,v);}
33     int query0(int a, int b, int i){ // ith element in range [a,b)
34         return query0(1,0,n,a,b,i); // (if it was sorted)

```

```

35     }
36     void upd(int i){ // swap positions i,i+1
37         upd(1,0,n,i);
38     }
39 };

```

2.8 STL extended set

```

1  #include<ext/pb_ds/assoc_container.hpp>
2  #include<ext/pb_ds/tree_policy.hpp>
3  using namespace __gnu_pbds;
4  typedef tree<int,null_type,less<int>,rb_tree_tag,
5              tree_order_statistics_node_update> ordered_set;
6  // find_by_order(i) -> iterator to ith element
7  // order_of_key(k) -> position (int) of lower_bound of k

```

2.9 STL rope

```

1  #include <ext/rope>
2  using namespace __gnu_cxx;
3  rope<int> s;
4  // Sequence with O(log(n)) random access, insert, erase at any position
5  // s.push_back(x);
6  // s.insert(i,r) // insert rope r at position i
7  // s.erase(i,k) // erase subsequence [i,i+k)
8  // s.substr(i,k) // return new rope corresponding to subsequence [i,i+k)
9  // s[i] // access ith element (cannot modify)
10 // s.mutable_reference_at(i) // acces ith element (allows modification)
11 // s.begin() and s.end() are const iterators (use mutable_begin(),
    mutable_end() to allow modification)

```

2.10 Treap (as BST)

```

1  typedef struct item *pitem;
2  struct item {
3      int pr,key,cnt;
4      pitem l,r;
5      item(int key):key(key),pr(rand()),cnt(1),l(0),r(0) {}
6  };
7  int cnt(pitem t){return t?t->cnt:0;}
8  void upd_cnt(pitem t){if(t)t->cnt=cnt(t->l)+cnt(t->r)+1;}
9  void split(pitem t, int key, pitem& l, pitem& r){ // l: < key, r: >= key
10     if(!t)l=r=0;
11     else if(key<t->key)split(t->l,key,l,t->l),r=t;

```

```

12 else split(t->r, key, t->r, r), l=t;
13 upd_cnt(t);
14 }
15 void insert(pitem& t, pitem it){
16     if(!t)t=it;
17     else if(it->pr>t->pr)split(t, it->key, it->l, it->r), t=it;
18     else insert(it->key<t->key?t->l:t->r, it);
19     upd_cnt(t);
20 }
21 void merge(pitem& t, pitem l, pitem r){
22     if(!l||!r)t=l?l:r;
23     else if(l->pr>r->pr)merge(l->r, l->r, r), t=l;
24     else merge(r->l, l, r->l), t=r;
25     upd_cnt(t);
26 }
27 void erase(pitem& t, int key){
28     if(t->key==key)merge(t, t->l, t->r);
29     else erase(key<t->key?t->l:t->r, key);
30     upd_cnt(t);
31 }
32 void unite(pitem& t, pitem l, pitem r){
33     if(!l||!r){t=l?l:r;return;}
34     if(l->pr<r->pr)swap(l, r);
35     pitem p1, p2; split(r, l->key, p1, p2);
36     unite(l->l, l->l, p1); unite(l->r, l->r, p2);
37     t=l; upd_cnt(t);
38 }
39 pitem kth(pitem t, int k){
40     if(!t)return 0;
41     if(k==cnt(t->l))return t;
42     return k<cnt(t->l)?kth(t->l, k):kth(t->r, k-cnt(t->l)-1);
43 }
44 pair<int, int> lb(pitem t, int key){ // position and value of lower_bound
45     if(!t)return {0, 1<<30}; // (special value)
46     if(key>t->key){
47         auto w=lb(t->r, key); w.first+=cnt(t->l)+1; return w;
48     }
49     auto w=lb(t->l, key);
50     if(w.first==cnt(t->l))w.second=t->key;
51     return w;
52 }

```

2.11 Treap (implicit key)

```

1 // example that supports range reverse and addition updates, and range
  // sum query
2 // (commented parts are specific to this problem)
3 typedef struct item *pitem;
4 struct item {
5     int pr, cnt, val;
6     // int sum; // (parameters for range query)
7     // bool rev; int add; // (parameters for lazy prop)
8     pitem l, r;
9     item(int val): pr(rand()), cnt(1), val(val), l(0), r(0) /*, sum(val), rev(0),
    // add(0) */ {}
10 };
11 void push(pitem it){
12     if(it){
13         /*if(it->rev){
14             swap(it->l, it->r);
15             if(it->l)it->l->rev^=true;
16             if(it->r)it->r->rev^=true;
17             it->rev=false;
18         }
19         it->val+=it->add; it->sum+=it->cnt*it->add;
20         if(it->l)it->l->add+=it->add;
21         if(it->r)it->r->add+=it->add;
22         it->add=0; */
23     }
24 }
25 int cnt(pitem t){return t?t->cnt:0;}
26 // int sum(pitem t){return t?push(t), t->sum:0;}
27 void upd_cnt(pitem t){
28     if(t){
29         t->cnt=cnt(t->l)+cnt(t->r)+1;
30         // t->sum=t->val+sum(t->l)+sum(t->r);
31     }
32 }
33 void merge(pitem& t, pitem l, pitem r){
34     push(l); push(r);
35     if(!l||!r)t=l?l:r;
36     else if(l->pr>r->pr)merge(l->r, l->r, r), t=l;
37     else merge(r->l, l, r->l), t=r;
38     upd_cnt(t);
39 }

```

```

40 void split(pitem t, pitem& l, pitem& r, int sz){ // sz:desired size of l
41     if(!t){l=r=0;return;}
42     push(t);
43     if(sz<=cnt(t->l))split(t->l,l,t->l,sz),r=t;
44     else split(t->r,t->r,r,sz-1-cnt(t->l)),l=t;
45     upd_cnt(t);
46 }
47 void output(pitem t){ // useful for debugging
48     if(!t)return;
49     push(t);
50     output(t->l);printf("_%d",t->val);output(t->r);
51 }
52 // use merge and split for range updates and queries

```

2.12 Treap (with node father)

```

1 // node father is useful to keep track of the chain of each node
2 // alternative: splay tree
3 // IMPORTANT: add pointer f in struct item
4 void merge(pitem& t, pitem l, pitem r){
5     push(l);push(r);
6     if(!l||!r)t=l?l:r;
7     else if(l->pr>r->pr)merge(l->r,l->r,r),l->r->f=t=l;
8     else merge(r->l,l,r->l),r->l->f=t=r;
9     upd_cnt(t);
10 }
11 void split(pitem t, pitem& l, pitem& r, int sz){
12     if(!t){l=r=0;return;}
13     push(t);
14     if(sz<=cnt(t->l)){
15         split(t->l,l,t->l,sz);r=t;
16         if(l)l->f=0;
17         if(t->l)t->l->f=t;
18     }
19     else {
20         split(t->r,t->r,r,sz-1-cnt(t->l));l=t;
21         if(r)r->f=0;
22         if(t->r)t->r->f=t;
23     }
24     upd_cnt(t);
25 }
26 void push_all(pitem t){
27     if(t->f)push_all(t->f);

```

```

28     push(t);
29 }
30 pitem root(pitem t, int& pos){ // get root and position for node t
31     push_all(t);
32     pos=cnt(t->l);
33     while(t->f){
34         pitem f=t->f;
35         if(t==f->r)pos+=cnt(f->l)+1;
36         t=f;
37     }
38     return t;
39 }

```

2.13 Link-Cut tree

```

1 typedef struct item *pitem;
2 struct item {
3     int pr;bool rev;
4     pitem l,r,f,d;
5     item():pr(rand()),l(0),r(0),f(0),d(0),rev(0){}
6 };
7 void push(pitem t){
8     if(t&&!--t->rev){
9         swap(t->l,t->r);
10        if(t->l)t->l->rev^=1;
11        if(t->r)t->r->rev^=1;
12        t->rev=0;
13    }
14 }
15 void merge(pitem& t, pitem l, pitem r){
16     push(l);push(r);
17     if(!l||!r)t=l?l:r;
18     else if(l->pr>r->pr)merge(l->r,l->r,r),l->r->f=t=l;
19     else merge(r->l,l,r->l),r->l->f=t=r;
20 }
21 void push_all(pitem t){
22     if(t->f)push_all(t->f);
23     push(t);
24 }
25 void split(pitem t, pitem& l, pitem& r){
26     push_all(t);
27     l=t->l;r=t->r;t->l=t->r=0;
28     while(t->f){

```

```

29 pitem f=t->f;t->f=0;
30 if(t==f->l){
31     if(r)r->f=f;
32     f->l=r;r=f;
33 }
34 else {
35     if(l)l->f=f;
36     f->r=l;l=f;
37 }
38 t=f;
39 }
40 if(l)l->f=0;
41 if(r)r->f=0;
42 }
43 pitem path(pitem p){return p->f?path(p->f):p;}
44 pitem tail(pitem p){push(p);return p->r?tail(p->r):p;}
45 pitem expose(pitem p){
46     pitem q,r,t;
47     split(p,q,r);
48     if(q)tail(q)->d=p;
49     merge(p,p,r);
50     while(t=tail(p),t->d){
51         pitem d=t->d;t->d=0;
52         split(d,q,r);
53         if(q)tail(q)->d=d;
54         merge(p,p,d);merge(p,p,r);
55     }
56     return p;
57 }
58 pitem root(pitem v){return tail(expose(v));}
59 void evert(pitem v){expose(v)->rev^=1;v->d=0;}
60 void link(pitem v, pitem w){ // make v son of w
61     evert(v);
62     pitem p=path(v);
63     merge(p,p,expose(w));
64 }
65 void cut(pitem v){ // cut v from its father
66     pitem p,q;
67     expose(v);split(v,p,q);v->d=0;
68 }
69 void cut(pitem v, pitem w){evert(w);cut(v);}

```

2.14 Convex hull trick (static)

```

1 typedef ll tc;
2 struct Line{tc m,h;};
3 struct CHT { // for minimum (for maximum just change the sign of lines)
4     vector<Line> c;
5     int pos=0;
6     tc in(Line a, Line b){
7         tc x=b.h-a.h,y=a.m-b.m;
8         return x/y+(x%y?!((x>0)^(y>0)):0); // ==ceil(x/y)
9     }
10 void add(tc m, tc h){ // m's should be non increasing
11     Line l=(Line){m,h};
12     if(c.size()&&m==c.back().m){
13         l.h=min(h,c.back().h);c.pop_back();if(pos)pos--;
14     }
15     while(c.size()>1&&in(c.back(),l)<=in(c[c.size()-2],c.back())){
16         c.pop_back();if(pos)pos--;
17     }
18     c.push_back(l);
19 }
20 inline bool fbin(tc x, int m){return in(c[m],c[m+1])>x;}
21 tc eval(tc x){
22     // O(log n) query:
23     int s=0,e=c.size();
24     while(e-s>1){int m=(s+e)/2;
25         if(fbin(x,m-1))e=m;
26         else s=m;
27     }
28     return c[s].m*x+c[s].h;
29     // O(1) query (for ordered x's):
30     while(pos>0&&fbin(x,pos-1))pos--;
31     while(pos<c.size()-1&&!fbin(x,pos))pos++;
32     return c[pos].m*x+c[pos].h;
33 }
34 };

```

2.15 Convex hull trick (dynamic)

```

1 typedef ll tc;
2 const tc is_query=-(1LL<<62); // special value for query
3 struct Line {
4     tc m,b;

```

```

5 mutable multiset<Line>::iterator it,end;
6 const Line* succ(multiset<Line>::iterator it) const {
7     return (++it==end? NULL : &*it);}
8 bool operator<(const Line& rhs) const {
9     if(rhs.b!=is_query)return m<rhs.m;
10    const Line *s=succ(it);
11    if(!s)return 0;
12    return b-s->b<(s->m-m)*rhs.m;
13 }
14 };
15 struct HullDynamic : public multiset<Line> { // for maximum
16     bool bad(iterator y){
17         iterator z=next(y);
18         if(y==begin()){
19             if(z==end())return false;
20             return y->m==z->m&&y->b<=z->b;
21         }
22         iterator x=prev(y);
23         if(z==end())return y->m==x->m&&y->b<=x->b;
24         return (x->b-y->b)*(z->m-y->m)>=(y->b-z->b)*(y->m-x->m);
25     }
26     iterator next(iterator y){return ++y;}
27     iterator prev(iterator y){return --y;}
28     void add(tc m, tc b){
29         iterator y=insert((Line){m,b});
30         y->it=y;y->end=end();
31         if(bad(y)){erase(y);return;}
32         while(next(y)!=end()&&bad(next(y)))erase(next(y));
33         while(y!=begin()&&bad(prev(y)))erase(prev(y));
34     }
35     tc eval(tc x){
36         Line l=*lower_bound((Line){x,is_query});
37         return l.m*x+l.b;
38     }
39 };

```

2.16 Li-Chao tree

```

1 // LiChaoTree lct(sz) -> allows queries for all x in range [0,sz)
2 // lct.add(L)         -> add line L to range [0,sz) in O(log(sz))
3 // lct.add(L,st,en)   -> add line L to range [st,en) in O(log^2(sz))
4 // lct.query(x)       -> query maximum L(x) across all current lines in
                        O(log(sz))

```

```

5
6 typedef long long T;
7 T INF=1e18;
8
9 struct Line{
10     T a, b;
11     Line():a(0),b(-INF){}
12     Line(T a, T b):a(a),b(b){}
13     T get(T x){return a*x+b;}
14 };
15
16 struct LiChaoTree{
17     struct Node{
18         Line line;
19         Node *lc,*rc=0;
20     }*root;
21     int sz;
22     LiChaoTree(int sz): root(new Node()), sz(sz){}
23
24     void addRec(Node* &n, int tl, int tr, Line x){
25         if(!n) n=new Node();
26         if(n->line.get(tl)<x.get(tl)) swap(n->line,x);
27         if(n->line.get(tr-1)>=x.get(tr-1)||tl+1==tr) return;
28         int mid=(tl+tr)/2;
29         if(n->line.get(mid)>x.get(mid)){
30             addRec(n->rc,mid,tr,x);
31         }
32         else{
33             swap(n->line,x);
34             addRec(n->lc,tl,mid,x);
35         }
36     }
37
38     void add(Node* &n, int tl, int tr, int l, int r, Line x){
39         if(tr<=l||r<=tl) return;
40         if(!n) n=new Node();
41         if(l<=tl&&tr<=r) return addRec(n,tl,tr,x);
42         int mid=(tl+tr)/2;
43         add(n->lc,tl,mid,l,r,x);
44         add(n->rc,mid,tr,l,r,x);
45     }
46
47     T query(Node* &n, int tl, int tr, int x){

```

```

48     if(!n) return -INF;
49     if(tl+1==tr) return n->line.get(x);
50     T res=n->line.get(x);
51     int mid=(tl+tr)/2;
52     if(x<mid) res=max(res, query(n->lc,tl,mid,x));
53     else res=max(res, query(n->rc,mid,tr,x));
54     return res;
55 }
56
57 void add(Line x){add(root,0,sz,0,sz,x);}
58 void add(Line x, int l, int r){add(root,0,sz,l,r,x);}
59 T query(int x){return query(root,0,sz,x);}
60 };

```

2.17 Gain-cost-set

```

1 // stores pairs (benefit,cost) (erases non-optimal pairs)
2 struct GCS {
3     set<pair<int,int> > s;
4     void add(int g, int c){
5         pair<int,int> x={g,c};
6         auto p=s.lower_bound(x);
7         if(p!=s.end()&&p->second<=x.second)return;
8         if(p!=s.begin()){ // erase pairs with less benefit
9             --p;           // and more cost
10            while(p->second>=x.second){
11                if(p==s.begin()){s.erase(p);break;}
12                s.erase(p--);
13            }
14        }
15        s.insert(x);
16    }
17    int get(int gain){ // min cost for some benefit
18        auto p=s.lower_bound((pair<int,int>){gain,-INF});
19        int r=p==s.end()?INF:p->second;
20        return r;
21    }
22 };

```

2.18 Disjoint intervals

```

1 // stores disjoint intervals as [first, second)
2 struct disjoint_intervals {
3     set<pair<int,int> > s;

```

```

4     void insert(pair<int,int> v){
5         if(v.first>=v.second) return;
6         auto at=s.lower_bound(v);auto it=at;
7         if(at!=s.begin()&&(--at)->second>=v.first)v.first=at->first,--it;
8         for(;it!=s.end()&&it->first<=v.second;s.erase(it++))
9             v.second=max(v.second,it->second);
10        segs.insert(v);
11    }
12 };

```

3 Graphs

3.1 Topological sort

```

1 vector<int> g[MAXN];int n;
2 vector<int> tsort(){ // lexicographically smallest topological sort
3     vector<int> r;priority_queue<int> q;
4     vector<int> d(2*n,0);
5     for (int i = 0; i < n; ++i)for (int j = 0; j < g[i].size(); ++j)d[g[i]
6         ][j]]++;
7     for (int i = 0; i < n; ++i)if(!d[i])q.push(-i);
8     while(!q.empty()){
9         int x=-q.top();q.pop();r.push_back(x);
10        for (int i = 0; i < g[x].size(); ++i){
11            d[g[x][i]]--;
12            if(!d[g[x][i]])q.push(-g[x][i]);
13        }
14    }
15    return r; // if not DAG it will have less than n elements
16 };

```

3.2 Kruskal (+ Union-Find)

```

1 int uf[MAXN];
2 void uf_init(){memset(uf,-1,sizeof(uf));}
3 int uf_find(int x){return uf[x]<0?x:uf[x]=uf_find(uf[x]);}
4 bool uf_join(int x, int y){
5     x=uf_find(x);y=uf_find(y);
6     if(x==y)return false;
7     if(uf[x]>uf[y])swap(x,y);
8     uf[x]+=uf[y];uf[y]=x;
9     return true;
10 };

```



```

11 vector<pair<ll,pair<int,int> > > es; // edges (cost,(u,v))
12 ll kruskal(){ // assumes graph is connected
13     sort(es.begin(),es.end());uf_init();
14     ll r=0;
15     for (int i = 0; i < es.size(); ++i){
16         int x=es[i].second.first,y=es[i].second.second;
17         if(uf_join(x,y))r+=es[i].first; // (x,y,c) belongs to mst
18     }
19     return r; // total cost
20 }

```

3.3 Dijkstra

```

1 vector<pair<int,int> > g[MAXN]; // u->[(v,cost)]
2 ll dist[MAXN];
3 void dijkstra(int x){
4     memset(dist,-1,sizeof(dist));
5     priority_queue<pair<ll,int> > q;
6     dist[x]=0;q.push({0,x});
7     while(!q.empty()){
8         x=q.top().second;ll c=-q.top().first;q.pop();
9         if(dist[x]!=c)continue;
10        for (int i = 0; i < g[x].size(); ++i){
11            int y=g[x][i].first; ll c=g[x][i].second;
12            if(dist[y]<0||dist[x]+c<dist[y])
13                dist[y]=dist[x]+c,q.push({-dist[y],y});
14        }
15    }
16 }

```

3.4 Bellman-Ford

```

1 int n;
2 vector<pair<int,int> > g[MAXN]; // u->[(v,cost)]
3 ll dist[MAXN];
4 void bford(int src){ // O(nm)
5     fill(dist,dist+n,INF);dist[src]=0;
6     for (int _ = 0; _ < n; ++_)for (int x = 0; x < n; ++x)if(dist[x]!=INF)
7         for(auto t:g[x]){
8             dist[t.first]=min(dist[t.first],dist[x]+t.second);
9         }
10    for (int x = 0; x < n; ++x)if(dist[x]!=INF)for(auto t:g[x]){
11        if(dist[t.first]>dist[x]+t.second){
12            // neg cycle: all nodes reachable from t.first have -INF distance

```

```

12 // to reconstruct neg cycle: save "prev" of each node, go up from
13 // t.first until repeating a node. this node and all nodes
14 // between the two occurrences form a neg cycle
15 }

```

3.5 Floyd-Warshall

```

1 // g[i][j]: weight of edge (i, j) or INF if there's no edge
2 // g[i][i]=0
3 ll g[MAXN][MAXN];int n;
4 void floyd(){ // O(n^3) . Replaces g with min distances
5     for (int k = 0; k < n; ++k)for (int i = 0; i < n; ++i)if(g[i][k]<INF)
6         for (int j = 0; j < n; ++j)if(g[k][j]<INF)
7             g[i][j]=min(g[i][j],g[i][k]+g[k][j]);
8 }
9 bool inNegCycle(int v){return g[v][v]<0;}
10 bool hasNegCycle(int a, int b){ // true iff there's neg cycle in between
11     for (int i = 0; i < n; ++i)if(g[a][i]<INF&&g[i][b]<INF&&g[i][i]<0)
12         return true;
13     return false;
14 }

```

3.6 Strongly connected components (+ 2-SAT)

```

1 // MAXN: max number of nodes or 2 * max number of variables (2SAT)
2 bool truth[MAXN]; // truth[cmp[i]]=value of variable i (2SAT)
3 int nvar;int neg(int x){return MAXN-1-x;} // (2SAT)
4 vector<int> g[MAXN];
5 int n,lw[MAXN],idx[MAXN],qidx,cmp[MAXN],qcmp;
6 stack<int> st;
7 void tjn(int u){
8     lw[u]=idx[u]==qidx;
9     st.push(u);cmp[u]=-2;
10    for(int v:g[u]){
11        if(!idx[v]||cmp[v]==-2){
12            if(!idx[v]) tjn(v);
13            lw[u]=min(lw[u],lw[v]);
14        }
15    }
16    if(lw[u]==idx[u]){
17        int x,l=-1;
18        do{x=st.top();st.pop();cmp[x]=qcmp;if(min(x,neg(x))<nvar)l=x;}

```



```

19     while(x!=u);
20     if(1!=-1)truth[qcmp]=(cmp[neg(1)]<0); // (2SAT)
21     qcmp++;
22 }
23 }
24 void scc(){
25     memset(idx,0,sizeof(idx));qidx=0;
26     memset(cmp,-1,sizeof(cmp));qcmp=0;
27     for (int i = 0; i < n; ++i)if(!idx[i])tjn(i);
28 }
29 // Only for 2SAT:
30 void addor(int a, int b){g[neg(a)].push_back(b);g[neg(b)].push_back(a);}
31 bool satisf(int _nvar){
32     nvar=_nvar;n=MAXN;scc();
33     for (int i = 0; i < nvar; ++i)if(cmp[i]==cmp[neg(i)])return false;
34     return true;
35 }

```

3.7 Articulation - Bridges - Biconnected

```

1 vector<int> g[MAXN];int n;
2 struct edge {int u,v,comp;bool bridge;};
3 vector<edge> e;
4 void add_edge(int u, int v){
5     g[u].push_back(e.size());g[v].push_back(e.size());
6     e.push_back((edge){u,v,-1,false});
7 }
8 int D[MAXN],B[MAXN],T;
9 int nbc; // number of biconnected components
10 int art[MAXN]; // articulation point iff !=0
11 stack<int> st; // only for biconnected
12 void dfs(int u,int pe){
13     B[u]=D[u]=T++;
14     for(int ne:g[u])if(ne!=pe){
15         int v=e[ne].u^e[ne].v^u;
16         if(D[v]<0){
17             st.push(ne);dfs(v,ne);
18             if(B[v]>D[u])e[ne].bridge = true; // bridge
19             if(B[v]>=D[u]){
20                 art[u]++; // articulation
21                 int last; // start biconnected
22                 do {
23                     last=st.top();st.pop();

```

```

24         e[last].comp=nbc;
25     } while(last!=ne);
26     nbc++; // end biconnected
27 }
28 B[u]=min(B[u],B[v]);
29 }
30 else if(D[v]<D[u])st.push(ne),B[u]=min(B[u],D[v]);
31 }
32 }
33 void doit(){
34     memset(D,-1,sizeof(D));memset(art,0,sizeof(art));
35     nbc=T=0;
36     for (int i = 0; i < n; ++i)if(D[i]<0)dfs(i,-1),art[i]--;
37 }

```

3.8 Chu-Liu (minimum spanning arborescence)

```

1 //O(n*m) minimum spanning tree in directed graph
2 //returns -1 if not possible
3 //included i-th edge if take[i]!=0
4 typedef int tw; tw INF=111<<30;
5 struct edge{int u,v,id;tw len;};
6 struct ChuLiu{
7     int n; vector<edge> e;
8     vector<int> inc,dec,take,pre,num,id,vis;
9     vector<tw> inw;
10    void add_edge(int x, int y, tw w){
11        inc.push_back(0); dec.push_back(0); take.push_back(0);
12        e.push_back({x,y,SZ(e),w});
13    }
14    ChuLiu(int n):n(n),pre(n),num(n),id(n),vis(n),inw(n){}
15    tw doit(int root){
16        auto e2=e;
17        tw ans=0; int eg=SZ(e)-1,pos=SZ(e)-1;
18        while(1){
19            for (int i = 0; i < n; ++i) inw[i]=INF,id[i]=vis[i]=-1;
20            for(auto ed:e2) if(ed.len<inw[ed.v]){
21                inw[ed.v]=ed.len; pre[ed.v]=ed.u;
22                num[ed.v]=ed.id;
23            }
24            inw[root]=0;
25            for (int i = 0; i < n; ++i) if(inw[i]==INF) return -1;
26            int tot=-1;

```

```

27     for (int i = 0; i < n; ++i){
28         ans+=inw[i];
29         if(i!=root)take[num[i]]++;
30         int j=i;
31         while(vis[j]!=i&&j!=root&&id[j]<0)vis[j]=i,j=pre[j];
32         if(j!=root&&id[j]<0){
33             id[j]=++tot;
34             for(int k=pre[j];k!=j;k=pre[k]) id[k]=tot;
35         }
36     }
37     if(tot<0)break;
38     for (int i = 0; i < n; ++i) if(id[i]<0)id[i]=++tot;
39     n=tot+1; int j=0;
40     for (int i = 0; i < SZ(e2); ++i){
41         int v=e2[i].v;
42         e2[j].v=id[e2[i].v];
43         e2[j].u=id[e2[i].u];
44         if(e2[j].v!=e2[j].u){
45             e2[j].len=e2[i].len-inw[v];
46             inc.push_back(e2[i].id);
47             dec.push_back(num[v]);
48             take.push_back(0);
49             e2[j++].id=++pos;
50         }
51     }
52     e2.resize(j);
53     root=id[root];
54 }
55 while(pos>eg){
56     if(take[pos]>0) take[inc[pos]]++, take[dec[pos]]--;
57     pos--;
58 }
59 return ans;
60 }
61 };

```

3.9 LCA - Binary Lifting

```

1 vector<int> g[1<<K];int n; // K such that 2^K>=n
2 int F[K][1<<K],D[1<<K];
3 void lca_dfs(int x){
4     for (int i = 0; i < g[x].size(); ++i){
5         int y=g[x][i];if(y==F[0][x])continue;

```

```

6         F[0][y]=x;D[y]=D[x]+1;lca_dfs(y);
7     }
8 }
9 void lca_init(){
10     D[0]=0;F[0][0]=-1;
11     lca_dfs(0);
12     for (int k = 1; k < K; ++k)for (int x = 0; x < n; ++x)
13         if(F[k-1][x]<0)F[k][x]=-1;
14         else F[k][x]=F[k-1][F[k-1][x]];
15 }
16 int lca(int x, int y){
17     if(D[x]<D[y])swap(x,y);
18     for(int k=K-1;k>=0;--k)if(D[x]-(1<<k)>=D[y])x=F[k][x];
19     if(x==y)return x;
20     for(int k=K-1;k>=0;--k)if(F[k][x]!=F[k][y])x=F[k][x],y=F[k][y];
21     return F[0][x];
22 }

```

3.10 Heavy-Light decomposition

```

1 vector<int> g[MAXN];
2 int wg[MAXN],dad[MAXN],dep[MAXN]; // weight,father,depth
3 void dfs1(int x){
4     wg[x]=1;
5     for(int y:g[x])if(y!=dad[x]){
6         dad[y]=x;dep[y]=dep[x]+1;dfs1(y);
7         wg[x]+=wg[y];
8     }
9 }
10 int curpos,pos[MAXN],head[MAXN];
11 void hld(int x, int c){
12     if(c<0)c=x;
13     pos[x]=curpos++;head[x]=c;
14     int mx=-1;
15     for(int y:g[x])if(y!=dad[x]&&(mx<0||wg[mx]<wg[y]))mx=y;
16     if(mx>=0)hld(mx,c);
17     for(int y:g[x])if(y!=mx&&y!=dad[x])hld(y,-1);
18 }
19 void hld_init(){dad[0]=-1;dep[0]=0;dfs1(0);curpos=0;hld(0,-1);}
20 int query(int x, int y, STree& rmq){
21     int r=NEUT;
22     while(head[x]!=head[y]){
23         if(dep[head[x]]>dep[head[y]])swap(x,y);

```

```

24     r=oper(r,rmq.query(pos[head[y]],pos[y]+1));
25     y=dad[head[y]];
26 }
27 if(dep[x]>dep[y])swap(x,y); // now x is lca
28 r=oper(r,rmq.query(pos[x],pos[y]+1));
29 return r;
30 }
31 // for updating: rmq.upd(pos[x],v);
32 // queries on edges: - assign values of edges to "child" node
33 //                   - change pos[x] to pos[x]+1 in query (line 28)

```

3.11 Centroid decomposition

```

1 vector<int> g[MAXN];int n;
2 bool tk[MAXN];
3 int fat[MAXN]; // father in centroid decomposition
4 int szt[MAXN]; // size of subtree
5 int calcsz(int x, int f){
6     szt[x]=1;
7     for(auto y:g[x])if(y!=f&&!tk[y])szt[x]+=calcsz(y,x);
8     return szt[x];
9 }
10 void cdfs(int x=0, int f=-1, int sz=-1){ // 0(nlogn)
11     if(sz<0)sz=calcsz(x,-1);
12     for(auto y:g[x])if(!tk[y]&&sz[y]*2>=sz){
13         szt[x]=0;cdfs(y,f,sz);return;
14     }
15     tk[x]=true;fat[x]=f;
16     for(auto y:g[x])if(!tk[y])cdfs(y,x);
17 }
18 void centroid(){memset(tk,false,sizeof(tk));cdfs();}

```

3.12 Parallel DFS

```

1 struct Tree {
2     int n,z[2];
3     vector<vector<int>> g;
4     vector<int> ex,ey,p,w,f,v[2];
5     Tree(int n):g(n),w(n),f(n){}
6     void add_edge(int x, int y){
7         p.push_back(g[x].size());g[x].push_back(ex.size());ex.push_back(x);
8         ey.push_back(y);
9         p.push_back(g[y].size());g[y].push_back(ex.size());ex.push_back(y);
10        ey.push_back(x);

```

```

9    }
10    bool go(int k){ // returns true if it finds new node
11        int& x=z[k];
12        while(x>0&&
13            (w[x]==g[x].size()||w[x]==g[x].size()-1&&(g[x].back()^1)==f[x]))
14            x=f[x]>0?ex[f[x]]:-1;
15        if(x<0)return false;
16        if((g[x][w[x]]^1)==f[x])w[x]++;
17        int e=g[x][w[x]],y=ey[e];
18        f[y]=e;w[x]++;w[y]=0;x=y;
19        v[k].push_back(x);
20        return true;
21    }
22    vector<int> erase_edge(int e){
23        e*=2; // erases eth edge, returns smaller component
24        int x=ex[e],y=ey[e];
25        p[g[x].back()]=p[e];
26        g[x][p[e]]=g[x].back();g[x].pop_back();
27        p[g[y].back()]=p[e^1];
28        g[y][p[e^1]]=g[y].back();g[y].pop_back();
29        f[x]=f[y]=-1;
30        w[x]=w[y]=0;
31        z[0]=x;z[1]=y;
32        v[0]={x};v[1]={y};
33        bool d0=true,d1=true;
34        while(d0&&d1)d0=go(0),d1=go(1);
35        if(d1)return v[0];
36        return v[1];
37    }
38 };

```

3.13 Eulerian path

```

1 // Directed version (uncomment commented code for undirected)
2 struct edge {
3     int y;
4     // list<edge>::iterator rev;
5     edge(int y):y(y){}
6 };
7 list<edge> g[MAXN];
8 void add_edge(int a, int b){
9     g[a].push_front(edge(b));//auto ia=g[a].begin();
10    // g[b].push_front(edge(a));auto ib=g[b].begin();

```

```

11 // ia->rev=ib;ib->rev=ia;
12 }
13 vector<int> p;
14 void go(int x){
15     while(g[x].size()){
16         int y=g[x].front().y;
17         //g[y].erase(g[x].front().rev);
18         g[x].pop_front();
19         go(y);
20     }
21     p.push_back(x);
22 }
23 vector<int> get_path(int x){ // get a path that begins in x
24 // check that a path exists from x before calling to get_path!
25     p.clear();go(x);reverse(p.begin(),p.end());
26     return p;
27 }

```

3.14 Dynamic connectivity

```

1 struct UnionFind {
2     int n,comp;
3     vector<int> uf,si,c;
4     UnionFind(int n=0):n(n),comp(n),uf(n),si(n,1){
5         for (int i = 0; i < n; ++i)uf[i]=i;}
6     int find(int x){return x==uf[x]?x:find(uf[x]);}
7     bool join(int x, int y){
8         if((x=find(x))==find(y))return false;
9         if(si[x]<si[y])swap(x,y);
10        si[x]+=si[y];uf[y]=x;comp--;c.push_back(y);
11        return true;
12    }
13    int snap(){return c.size();}
14    void rollback(int snap){
15        while(c.size()>snap){
16            int x=c.back();c.pop_back();
17            si[uf[x]]-=si[x];uf[x]=x;comp++;
18        }
19    }
20 };
21 enum {ADD,DEL,QUERY};
22 struct Query {int type,x,y};
23 struct DynCon {

```

```

24     vector<Query> q;
25     UnionFind dsu;
26     vector<int> mt;
27     map<pair<int,int>,int> last;
28     DynCon(int n):dsu(n){}
29     void add(int x, int y){
30         if(x>y)swap(x,y);
31         q.push_back((Query){ADD,x,y});mt.push_back(-1);last[{x,y}]=q.size()-1;
32     }
33     void remove(int x, int y){
34         if(x>y)swap(x,y);
35         q.push_back((Query){DEL,x,y});
36         int pr=last[{x,y}];mt[pr]=q.size()-1;mt.push_back(pr);
37     }
38     void query(){q.push_back((Query){QUERY,-1,-1});mt.push_back(-1);}
39     void process(){ // answers all queries in order
40         if(!q.size())return;
41         for (int i = 0; i < q.size(); ++i)if(q[i].type==ADD&&mt[i]<0)mt[i]=q[i].size()-1;
42         go(0,q.size());
43     }
44     void go(int s, int e){
45         if(s+1==e){
46             if(q[s].type==QUERY) // answer query using DSU
47                 printf("%d\n",dsu.comp);
48             return;
49         }
50         int k=dsu.snap(),m=(s+e)/2;
51         for(int i=e-1;i>=m;--i)if(mt[i]>=0&&mt[i]<s)dsu.join(q[i].x,q[i].y);
52         go(s,m);dsu.rollback(k);
53         for(int i=m-1;i>=s;--i)if(mt[i]>=e)dsu.join(q[i].x,q[i].y);
54         go(m,e);dsu.rollback(k);
55     }
56 };

```

3.15 Dominator tree

```

1 //idom[i]=parent of i in dominator tree with root=rt, or -1 if not exists
2 int n,rnk[MAXN],pre[MAXN],anc[MAXN],idom[MAXN],semi[MAXN],low[MAXN];
3 vector<int> g[MAXN],rev[MAXN],dom[MAXN],ord;
4 void dfspre(int pos){

```

```

5   rnk[pos]=SZ(ord); ord.push_back(pos);
6   for(auto x:g[pos]){
7       rev[x].push_back(pos);
8       if(rnk[x]==n) pre[x]=pos,dfspre(x);
9   }
10  }
11  int eval(int v){
12      if(anc[v]<n&&anc[anc[v]]<n){
13          int x=eval(anc[v]);
14          if(rnk[semi[low[v]]]>rnk[semi[x]]) low[v]=x;
15          anc[v]=anc[anc[v]];
16      }
17      return low[v];
18  }
19  void dominators(int rt){
20      for (int i = 0; i < n; ++i){
21          dom[i].clear(); rev[i].clear();
22          rnk[i]=pre[i]=anc[i]=idom[i]=n;
23          semi[i]=low[i]=i;
24      }
25      ord.clear(); dfspre(rt);
26      for(int i=SZ(ord)-1;i;i--){
27          int w=ord[i];
28          for(int v:rev[w]){
29              int u=eval(v);
30              if(rnk[semi[w]]>rnk[semi[u]])semi[w]=semi[u];
31          }
32          dom[semi[w]].push_back(w); anc[w]=pre[w];
33          for(int v:dom[pre[w]]){
34              int u=eval(v);
35              idom[v]=(rnk[pre[w]]>rnk[semi[u]]?u:pre[w]);
36          }
37          dom[pre[w]].clear();
38      }
39      for(int w:ord) if(w!=rt&&idom[w]!=semi[w]) idom[w]=idom[idom[w]];
40      for (int i = 0; i < n; ++i) if(idom[i]==n)idom[i]=-1;
41  }

```

3.16 Edmond's blossom (matching in general graphs)

```

1  vector<int> g[MAXN];
2  int n,m,mt[MAXN],qh,qt,q[MAXN],ft[MAXN],bs[MAXN];
3  bool inq[MAXN],inb[MAXN],inp[MAXN];

```

```

4  int lca(int root, int x, int y){
5      memset(inp,0,sizeof(inp));
6      while(1){
7          inp[x=bs[x]]=true;
8          if(x==root)break;
9          x=ft[mt[x]];
10     }
11     while(1){
12         if(inp[y=bs[y]])return y;
13         else y=ft[mt[y]];
14     }
15 }
16 void mark(int z, int x){
17     while(bs[x]!=z){
18         int y=mt[x];
19         inb[bs[x]]=inb[bs[y]]=true;
20         x=ft[y];
21         if(bs[x]!=z)ft[x]=y;
22     }
23 }
24 void contr(int s, int x, int y){
25     int z=lca(s,x,y);
26     memset(inb,0,sizeof(inb));
27     mark(z,x);mark(z,y);
28     if(bs[x]!=z)ft[x]=y;
29     if(bs[y]!=z)ft[y]=x;
30     for (int x = 0; x < n; ++x)if(inb[bs[x]]){
31         bs[x]=z;
32         if(!inq[x])inq[q[++qt]=x]=true;
33     }
34 }
35 int findp(int s){
36     memset(inq,0,sizeof(inq));
37     memset(ft,-1,sizeof(ft));
38     for (int i = 0; i < n; ++i)bs[i]=i;
39     inq[q[qh=qt=0]=s]=true;
40     while(qh<=qt){
41         int x=q[qh++];
42         for(int y:g[x])if(bs[x]!=bs[y]&&mt[x]!=y){
43             if(y==s|mt[y]>=0&&ft[mt[y]]>=0)contr(s,x,y);
44             else if(ft[y]<0){
45                 ft[y]=x;
46                 if(mt[y]<0)return y;

```

```

47     else if(!inq[mt[y]])inq[q[++qt]=mt[y]]=true;
48     }
49     }
50     }
51     return -1;
52 }
53 int aug(int s, int t){
54     int x=t,y,z;
55     while(x>=0){
56         y=ft[x];
57         z=mt[y];
58         mt[y]=x;mt[x]=y;
59         x=z;
60     }
61     return t>=0;
62 }
63 int edmonds(){ // O(n^2 m)
64     int r=0;
65     memset(mt,-1,sizeof(mt));
66     for (int x = 0; x < n; ++x)if(mt[x]<0)r+=aug(x,findp(x));
67     return r;
68 }

```

4 Math

4.1 Identities

$$C_n = \frac{2(2n-1)}{n+1} C_{n-1}$$

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

$$C_n \sim \frac{4^n}{n^{3/2} \sqrt{\pi}}$$

$\sigma(n) = O(\log(\log(n)))$ (number of divisors of n)

$$F_{2n+1} = F_n^2 + F_{n+1}^2$$

$$F_{2n} = F_{n+1}^2 - F_{n-1}^2$$

$$\sum_{i=1}^n F_i = F_{n+2} - 1$$

$$F_{n+i}F_{n+j} - F_nF_{n+i+j} = (-1)^n F_i F_j$$

(Möbius Inv. Formula) Let $g(n) = \sum_{d|n} f(d)$, then $f(n) = \sum_{d|n} d \mu\left(\frac{n}{d}\right) g\left(\frac{n}{d}\right)$.

4.2 Theorems

- 1 (Tutte) A graph, $G = (V, E)$, has a perfect matching if and only if for every subset U of V , the subgraph induced by $V - U$ has at most $|U|$ connected components with an odd number of vertices.

- 2 Petersens Theorem. Every cubic, bridgeless graph contains a perfect matching.
- 3 (Dilworth) In any finite partially ordered set, the maximum number of elements in any antichain equals the minimum number of chains in any partition of the set into chains
- 4 Pick: $A = I + B/2 - 1$ (area of polygon, points inside, points on border)

4.3 Integer floor division

```

1 void floordiv(ll x, ll y, ll& q, ll& r) { // (for negative x)
2     q=x/y;r=x%y;
3     if((r!=0)&&((r<0)!=(y<0)))q--,r+=y;
4 }

```

4.4 Sieve of Eratosthenes

```

1 int cr[MAXN]; // -1 if prime, some not trivial divisor if not
2 void init_sieve(){
3     memset(cr,-1,sizeof(cr));
4     for (int i = 2; i < MAXN; ++i)if(cr[i]<0)for(ll j=1LL*i*i;j<MAXN;j+=i)
5         cr[j]=i;
6 }
7 map<int,int> fact(int n){ // must call init_cribe before
8     map<int,int> r;
9     while(cr[n]>=0)r[cr[n]]++,n/=cr[n];
10    if(n>1)r[n]++;
11    return r;

```

4.5 Generate divisors

```

1 void div_rec(vector<ll>& r, vector<pair<ll,int> >& f, int k, ll c){
2     if(k==f.size()){r.push_back(c);return;}
3     for (int i = 0; i < f[k].second+1; ++i)div_rec(r,f,k+1,c),c*=f[k].first;
4 }
5 vector<ll> divisors(vector<pair<ll,int> > f){
6     vector<ll> r; // returns divisors given factorization
7     div_rec(r,f,0,1);
8     return r;
9 }

```

4.6 Pollard's rho

```

1 ll gcd(ll a, ll b){return a?gcd(b%a,a):b;}
2 ll mulmod(ll a, ll b, ll m) {
3     ll r=a*b-(ll)((long double)a*b/m+.5)*m;
4     return r<0?r+m:r;
5 }
6 ll expmod(ll b, ll e, ll m){
7     if(!e)return 1;
8     ll q=expmod(b,e/2,m);q=mulmod(q,q,m);
9     return e&1?mulmod(b,q,m):q;
10 }
11 bool is_prime_prob(ll n, int a){
12     if(n==a)return true;
13     ll s=0,d=n-1;
14     while(d%2==0)s++,d/=2;
15     ll x=expmod(a,d,n);
16     if((x==1)|| (x+1==n))return true;
17     for (int _ = 0; _ < s-1; ++_){
18         x=mulmod(x,x,n);
19         if(x==1)return false;
20         if(x+1==n)return true;
21     }
22     return false;
23 }
24 bool rabin(ll n){ // true iff n is prime
25     if(n==1)return false;
26     int ar[]={2,3,5,7,11,13,17,19,23};
27     for (int i = 0; i < 9; ++i)if(!is_prime_prob(n,ar[i]))return false;
28     return true;
29 }
30 ll rho(ll n){
31     if(!(n&1))return 2;
32     ll x=2,y=2,d=1;
33     ll c=rand()%n+1;
34     while(d==1){
35         x=(mulmod(x,x,n)+c)%n;
36         y=(mulmod(y,y,n)+c)%n;
37         y=(mulmod(y,y,n)+c)%n;
38         if(x>=y)d=gcd(x-y,n);
39         else d=gcd(y-x,n);
40     }
41     return d==n?rho(n):d;
42 }
43 void fact(ll n, map<ll,int>& f){ //O (lg n)^3

```

```

44     if(n==1)return;
45     if(rabin(n)){f[n]++;return;}
46     ll q=rho(n);fact(q,f);fact(n/q,f);
47 }
48 // optimized version: replace rho and fact with the following:
49 const int MAXP=1e6+1; // sieve size
50 int sv[MAXP]; // sieve
51 ll add(ll a, ll b, ll m){return (a+=b)<m?a-a-m;}
52 ll rho(ll n){
53     static ll s[MAXP];
54     while(1){
55         ll x=rand()%n,y=x,c=rand()%n;
56         ll *px=s,*py=s,v=0,p=1;
57         while(1){
58             *py++=y=add(mulmod(y,y,n),c,n);
59             *py++=y=add(mulmod(y,y,n),c,n);
60             if((x=*px++)==y)break;
61             ll t=p;
62             p=mulmod(p,abs(y-x),n);
63             if(!p)return gcd(t,n);
64             if(++v==26){
65                 if((p=gcd(p,n))>1&&p<n)return p;
66                 v=0;
67             }
68         }
69         if(v&&(p=gcd(p,n))>1&&p<n)return p;
70     }
71 }
72 void init_sv(){
73     for (int i = 2; i < MAXP; ++i)if(!sv[i])for(ll j=i;j<MAXP;j+=i)sv[j]=i;
74 }
75 void fact(ll n, map<ll,int>& f){ // call init_sv first!!!
76     for(auto&& p:f){
77         while(n%p.first==0){
78             p.second++;
79             n/=p.first;
80         }
81     }
82     if(n<MAXP)while(n>1)f[sv[n]]++,n/=sv[n];
83     else if(rabin(n))f[n]++;
84     else {ll q=rho(n);fact(q,f);fact(n/q,f);}
85 }

```


4.7 Simpson's rule

```

1 double integrate(double f(double), double a, double b, int n=10000){
2     double r=0,h=(b-a)/n,fa=f(a),fb;
3     for (int i = 0; i < n; ++i){fb=f(a+h*(i+1));r+=fa+4*f(a+h*(i+0.5))+fb;
4         fa=fb;}
5     return r*h/6.;
6 }

```

4.8 Polynomials

```

1 typedef int tp; // type of polynomial
2 template<class T=tp>
3 struct poly { // poly<> : 1 variable, poly<poly<>>: 2 variables, etc.
4     vector<T> c;
5     T& operator[](int k){return c[k];}
6     poly(vector<T>& c):c(c){}
7     poly(initializer_list<T> c):c(c){}
8     poly(int k):c(k){}
9     poly(){}
10    poly operator+(poly<T> o){
11        int m=c.size(),n=o.c.size();
12        poly res(max(m,n));
13        for (int i = 0; i < m; ++i)res[i]=res[i]+c[i];
14        for (int i = 0; i < n; ++i)res[i]=res[i]+o.c[i];
15        return res;
16    }
17    poly operator*(tp k){
18        poly res(c.size());
19        for (int i = 0; i < c.size(); ++i)res[i]=c[i]*k;
20        return res;
21    }
22    poly operator*(poly o){
23        int m=c.size(),n=o.c.size();
24        poly res(m+n-1);
25        for (int i = 0; i < m; ++i)for (int j = 0; j < n; ++j)res[i+j]=res[i+j]+c[i]*o.c[j];
26        return res;
27    }
28    poly operator-(poly<T> o){return *this+(o*-1);}
29    T operator()(tp v){
30        T sum(0);
31        for(int i=c.size()-1;i>=0;--i)sum=sum*v+c[i];

```

```

32     return sum;
33 }
34 };
35 // example: p(x,y)=2*x^2+3*x*y-y+4
36 // poly<poly<>> p={{4,-1},{0,3},{2}}
37 // printf("%d\n",p(2)(3)) // 27 (p(2,3))
38 set<tp> roots(poly<> p){ // only for integer polynomials
39     set<tp> r;
40     while(!p.c.empty()&&!p.c.back())p.c.pop_back();
41     if(!p(0))r.insert(0);
42     if(p.c.empty())return r;
43     tp a0=0,an=abs(p[p.c.size()-1]);
44     for(int k=0;!a0;a0=abs(p[k++]));
45     vector<tp> ps,qs;
46     for (int i = 1; i < sqrt(a0)+1; ++i)if(a0%i==0)ps.push_back(i),ps.
47         push_back(a0/i);
48     for (int i = 1; i < sqrt(an)+1; ++i)if(an%i==0)qs.push_back(i),qs.
49         push_back(an/i);
50     for(auto pt:ps)for(auto qt:qs)if(pt%qt==0){
51         tp x=pt/qt;
52         if(!p(x))r.insert(x);
53         if(!p(-x))r.insert(-x);
54     }
55     return r;
56 }
57 pair<poly<>,tp> ruffini(poly<> p, tp r){ // returns pair (result,rem)
58     int n=p.c.size()-1;
59     vector<tp> b(n);
60     b[n-1]=p[n];
61     for(int k=n-2;k>=0;--k)b[k]=p[k+1]+r*b[k+1];
62     return {poly<>(b),p[0]+r*b[0]};
63 }
64 // only for double polynomials
65 pair<poly<>,poly<>> polydiv(poly<> p, poly<> q){ // returns pair (
66     result,rem)
67     int n=p.c.size()-q.c.size()+1;
68     vector<tp> b(n);
69     for(int k=n-1;k>=0;--k){
70         b[k]=p.c.back()/q.c.back();
71         for (int i = 0; i < q.c.size(); ++i)p[i+k]-=b[k]*q[i];
72         p.c.pop_back();
73     }
74     while(!p.c.empty()&&abs(p.c.back())<EPS)p.c.pop_back();

```



```

72     return {poly<>(b),p};
73 }
74 // only for double polynomials
75 poly<> interpolate(vector<tp> x, vector<tp> y){ //TODO TEST
76     poly<> q={1},S={0};
77     for(tp a:x)q=poly<>({-a,1})*q;
78     for (int i = 0; i < x.size(); ++i){
79         poly<> Li=ruffini(q,x[i]).first;
80         Li=Li*(1.0/Li(x[i])); // change for int polynomials
81         S=S+Li*y[i];
82     }
83     return S;
84 }

```

4.9 Bairstow

```

1 double pget(poly<>& p, int k){return k<p.c.size()?p[k]:0;}
2 poly<> bairstow(poly<> p){ // returns polynomial of degree 2 that
3     int n=p.c.size()-1; // divides p
4     assert(n>=3&&abs(p.c.back())>EPS);
5     double u=p[n-1]/p[n],v=p[n-2]/p[n];
6     for (int _ = 0; _ < ITER; ++_){
7         auto w=polydiv(p,{v,u,1});
8         poly<> q=w.first,r0=w.second;
9         poly<> r1=polydiv(q,{v,u,1}).second;
10        double c=pget(r0,1),d=pget(r0,0),g=pget(r1,1),h=pget(r1,0);
11        double det=1/(v*g*g+h*(h-u*g)),uu=u;
12        u-=det*(-h*c+g*d);v-=det*(-g*v*c+(g*uu-h)*d);
13    }
14    return {v,u,1};
15 }
16 void addr(vector<double>& r, poly<>& p){
17     assert(p.c.size()<=3);
18     if(p.c.size()<=1)return;
19     if(p.c.size()==2)r.push_back(-p[0]/p[1]);
20     if(p.c.size()==3){
21         double a=p[2],b=p[1],c=p[0];
22         double d=b*b-4*a*c;
23         if(d<-0.1)return; // huge epsilon because of bad precision
24         d=d>0?sqrt(d):0;r.push_back((-b-d)/2/a);r.push_back((-b+d)/2/a);
25     }
26 }
27 }

```

```

28 vector<double> roots(poly<> p){
29     while(!p.c.empty()&&abs(p.c.back())<EPS)p.c.pop_back();
30     for (int i = 0; i < p.c.size(); ++i)p[i]/=p.c.back();
31     vector<double> r;int n;
32     while((n=p.c.size()-1)>=3){
33         poly<> q=bairstow(p);addr(r,q);
34         p=polydiv(p,q).first;
35         while(p.c.size()>n-1)p.c.pop_back();
36     }
37     addr(r,p);
38     return r;
39 }

```

4.10 Fast Fourier Transform

```

1 // MAXN must be power of 2 !!
2 // MOD-1 needs to be a multiple of MAXN !!
3 // big mod and primitive root for NTT:
4 typedef ll tf;
5 typedef vector<tf> poly;
6 const tf MOD=2305843009255636993,RT=5;
7 // FFT
8 struct CD {
9     double r,i;
10    CD(double r=0, double i=0):r(r),i(i){}
11    double real()const{return r;}
12    void operator/=(const int c){r/=c, i/=c;}
13 };
14 CD operator*(const CD& a, const CD& b){
15     return CD(a.r*b.r-a.i*b.i,a.r*b.i+a.i*b.r);}
16 CD operator+(const CD& a, const CD& b){return CD(a.r+b.r,a.i+b.i);}
17 CD operator-(const CD& a, const CD& b){return CD(a.r-b.r,a.i-b.i);}
18 const double pi=acos(-1.0);
19 // NTT
20 /*
21 struct CD {
22     tf x;
23     CD(tf x):x(x){}
24     CD(){}
25 };
26 CD operator*(const CD& a, const CD& b){return CD(mulmod(a.x,b.x));}
27 CD operator+(const CD& a, const CD& b){return CD(addmod(a.x,b.x));}
28 CD operator-(const CD& a, const CD& b){return CD(submod(a.x,b.x));}

```

```

29 vector<tf> rts(MAXN+9,-1);
30 CD root(int n, bool inv){
31     tf r=rts[n]<0?rts[n]=pm(RT,(MOD-1)/n):rts[n];
32     return CD(inv?pm(r,MOD-2):r);
33 }
34 */
35 CD cp1[MAXN+9],cp2[MAXN+9];
36 int R[MAXN+9];
37 void dft(CD* a, int n, bool inv){
38     for (int i = 0; i < n; ++i)if(R[i]<i)swap(a[R[i]],a[i]);
39     for(int m=2;m<=n;m*=2){
40         double z=2*pi/m*(inv?-1:1); // FFT
41         CD wi=CD(cos(z),sin(z)); // FFT
42         // CD wi=root(m,inv); // NTT
43         for(int j=0;j<n;j+=m){
44             CD w(1);
45             for(int k=j,k2=j+m/2;k2<j+m;k++,k2++){
46                 CD u=a[k];CD v=a[k2]*w;a[k]=u+v;a[k2]=u-v;w=w*wi;
47             }
48         }
49     }
50     if(inv)for (int i = 0; i < n; ++i)a[i]/=n; // FFT
51     //if(inv){ // NTT
52     //    CD z(pm(n,MOD-2)); // pm: modular exponentiation
53     //    for (int i = 0; i < n; ++i)a[i]=a[i]*z;
54     //}
55 }
56 poly multiply(poly& p1, poly& p2){
57     int n=p1.size()+p2.size()+1;
58     int m=1,cnt=0;
59     while(m<=n)m+=m,cnt++;
60     for (int i = 0; i < m; ++i){R[i]=0;for (int j = 0; j < cnt; ++j)R[i]=(
        R[i]<<1)|((i>>j)&1);}
61     for (int i = 0; i < m; ++i)cp1[i]=0,cp2[i]=0;
62     for (int i = 0; i < p1.size(); ++i)cp1[i]=p1[i];
63     for (int i = 0; i < p2.size(); ++i)cp2[i]=p2[i];
64     dft(cp1,m,false);dft(cp2,m,false);
65     for (int i = 0; i < m; ++i)cp1[i]=cp1[i]*cp2[i];
66     dft(cp1,m,true);
67     poly res;
68     n-=2;
69     for (int i = 0; i < n; ++i)res.push_back((tf)floor(cp1[i].real()+0.5))
        ; // FFT

```

```

70 //for (int i = 0; i < n; ++i)res.push_back(cp1[i].x); // NTT
71 return res;
72 }

```

4.11 FFT operations

```

1 // Polynomial division: O(n*log(n))
2 // Multi-point polynomial evaluation for arbitrarily large points: O(n*
    log^2(n))
3 // Multi-point polynomial evaluation for all points in [0, MOD): O((n+
    MOD)*log(n+MOD))
4 // Polynomial interpolation: O(n*log^2(n))
5 // Inverse: O(n*log(n))
6 // Log: O(n*log(n))
7 // Exp: O(n*log(n))
8
9 //Works with NTT. For FFT, just replace addmod,submod,mulmod,inv
10 poly add(poly &a, poly &b){
11     int n=SZ(a),m=SZ(b);
12     poly ans(max(n,m));
13     for (int i = 0; i < max(n,m); ++i) ans[i]=addmod(i<SZ(a)?a[i]:0, i<SZ(
        b)?b[i]:0);
14     while(SZ(ans)>1&&!ans.back())ans.pop_back();
15     return ans;
16 }
17
18 // derivative of p
19 poly derivate(poly &p){
20     poly ans(max(1, SZ(p)-1));
21     for (int i = 1; i < SZ(p); ++i) ans[i-1]=mulmod(p[i],i);
22     return ans;
23 }
24
25 // integral of p
26 poly integrate(poly &p){
27     poly ans(SZ(p)+1);
28     for (int i = 0; i < SZ(p); ++i) ans[i+1]=mulmod(p[i], inv(i+1));
29     return ans;
30 }
31
32 // p % (x^n)
33 poly takemod(poly &p, int n){
34     poly res=p;

```

```

35     res.resize(min(SZ(res),n));
36     while(SZ(res)>1&&res.back()==0) res.pop_back();
37     return res;
38 }
39
40 // first d terms of 1/p
41 poly invert(poly &p, int d){
42     assert(p[0]);
43     poly res={inv(p[0])};
44     int sz=1;
45     while(sz<d){
46         sz*=2;
47         poly pre(p.begin(), p.begin()+min(SZ(p),sz));
48         poly cur=multiply(res,pre);
49         for (int i = 0; i < SZ(cur); ++i) cur[i]=submod(0,cur[i]);
50         cur[0]=addmod(cur[0],2);
51         res=multiply(res,cur);
52         res=takemod(res,sz);
53     }
54     res.resize(d);
55     return res;
56 }
57
58 // first d terms of log(p)
59 poly log(poly &p, int d){
60     assert(p[0]==1);
61     poly cur=takemod(p,d);
62     poly a=invert(cur,d), b=derivate(cur);
63     auto res=multiply(a,b);
64     res=takemod(res,d-1);
65     res=integrate(res);
66     return res;
67 }
68
69 // first d terms of exp(p)
70 poly exp(poly &p, int d){
71     assert(!p[0]);
72     poly res={1};
73     int sz=1;
74     while(sz<d){
75         sz*=2;
76         poly lg=log(res, sz), cur(sz);
77         for (int i = 0; i < sz; ++i) cur[i]=submod(i<SZ(p)?p[i]:0, i<SZ(lg)?

```

```

        lg[i]:0);
78         cur[0]=addmod(cur[0],1);
79         res=multiply(res,cur);
80         res=takemod(res, sz);
81     }
82
83     res.resize(d);
84     return res;
85 }
86
87 pair<poly,poly> divslow(poly &a, poly &b){
88     poly q,r=a;
89     while(SZ(r)>=SZ(b)){
90         q.push_back(mulmod(r.back(),inv(b.back())));
91         if(q.back()) for (int i = 0; i < SZ(b); ++i){
92             r[SZ(r)-i-1]=submod(r[SZ(r)-i-1],mulmod(q.back(),b[SZ(b)-i-1]));
93         }
94         r.pop_back();
95     }
96     reverse(ALL(q));
97     return {q,r};
98 }
99
100 pair<poly,poly> divide(poly &a, poly &b){ //returns {quotient,remainder}
101     int m=SZ(a),n=SZ(b),MAGIC=750;
102     if(m<n) return {{0},a};
103     if(min(m-n,n)<MAGIC)return divslow(a,b);
104     poly ap=a; reverse(ALL(ap));
105     poly bp=b; reverse(ALL(bp));
106     bp=invert(bp,m-n+1);
107     poly q=multiply(ap,bp);
108     q.resize(SZ(q)+m-n-SZ(q)+1,0);
109     reverse(ALL(q));
110     poly bq=multiply(b,q);
111     for (int i = 0; i < SZ(bq); ++i) bq[i]=submod(0,bq[i]);
112     poly r=add(a,bq);
113     return {q,r};
114 }
115
116 vector<poly> tree;
117
118 void filltree(vector<tf> &x){
119     int k=SZ(x);

```

```

120 tree.resize(2*k);
121 for (int i = k; i < 2*k; ++i) tree[i]={submod(0,x[i-k]),1};
122 for(int i=k-1;i;--i) tree[i]=multiply(tree[2*i],tree[2*i+1]);
123 }
124
125 // Multi-point polynomial evaluation for arbitrarily large points
126 vector<tf> evaluate(poly &a, vector<tf> &x){
127     filltree(x);
128     int k=SZ(x);
129     vector<poly> ans(2*k);
130     ans[1]=divide(a,tree[1]).second;
131     for (int i = 2; i < 2*k; ++i) ans[i]=divide(ans[i>>1],tree[i]).second;
132     vector<tf> r; for (int i = 0; i < k; ++i) r.push_back(ans[i+k][0]);
133     return r;
134 }
135
136 // Given any number g, perform multi-point polynomial evaluation
137 // for all points of the form g^i, for each i in [0, k)
138 // in O((n+k) * log(n+k))
139 vector<int> chirpTransform(poly& p, int g, int k) {
140     int inv_g=inv(g), n=SZ(p)-1, sz=min(k,MOD-1), gk=1;
141     poly ap(n+1), bp(n+sz);
142
143     for (int i = 0; i < n+sz; ++i){
144         ll exp=1ll*i*(i-1)/2%(MOD-1);
145         if(i<=n) ap[n-i]=mulmod(p[i],fpow(inv_g, exp));
146         bp[i]=fpow(g, exp);
147     }
148
149     poly cp=multiply(ap, bp);
150     vector<int> ans(k);
151
152     for (int i = 0; i < sz; ++i){
153         ll exp=1ll*i*(i-1)/2%(MOD-1);
154         int val=0;
155         if(n+i<SZ(cp)) val=cp[n+i];
156         val=mulmod(val, fpow(inv_g,exp));
157         ans[i]=val;
158         gk=mulmod(gk,g);
159     }
160     for (int i = sz; i < k; ++i) ans[i]=ans[i-MOD+1];
161     return ans;
162 }

```

```

163
164 int get_primitive_root(int p) {
165     if(p==2)return 1;
166     int phi=p-1, n=phi;
167     vector<int> fact;
168     for(int i=2;i*i<=n;i++) if(n%i==0){
169         fact.push_back(i);
170         while(n%i==0) n/=i;
171     }
172     if(n>1) fact.push_back(n);
173
174     for (int res = 2; res < p+1; ++res){
175         int ok=1;
176         for(int f:fact){
177             if(fpow(res,phi/f)==1){
178                 ok=false;
179                 break;
180             }
181         }
182         if(ok)return res;
183     }
184     return -1;
185 }
186
187 // Multi-point polynomial evaluation for all points in [0, MOD)
188 // MOD needs to be a prime number
189 vector<int> evaluate_all_points(poly& p) {
190     int g=get_primitive_root(MOD), gk=1;
191     assert(g!=-1);
192
193     vector<int> ch=chirpTransform(p,g,MOD-1), ans(MOD, p[0]);
194
195     for (int i = 0; i < MOD-1; ++i){
196         ans[gk]=ch[i];
197         gk=mulmod(gk,g);
198     }
199     return ans;
200 }
201
202 poly interpolate(vector<tf> &x, vector<tf> &y){
203     filltree(x);
204     poly p=derivate(tree[1]);
205     int k=SZ(y);

```

```

206 vector<tf> d=evaluate(p,x);
207 vector<poly> intree(2*k);
208 for (int i = k; i < 2*k; ++i) intree[i]={mulmod(y[i-k],inv(d[i-k]))};
209 for(int i=k-1;i;i--){
210     poly p1=multiply(tree[2*i],intree[2*i+1]);
211     poly p2=multiply(tree[2*i+1],intree[2*i]);
212     intree[i]=add(p1,p2);
213 }
214 return intree[1];
215 }

```

4.12 Fast Hadamard Transform

```

1 ll c1[MAXN+9],c2[MAXN+9]; // MAXN must be power of 2 !!
2 void fht(ll* p, int n, bool inv){
3     for(int l=1;2*l<=n;l*=2)for(int i=0;i<n;i+=2*l)for (int j = 0; j < l;
4         ++j){
5         ll u=p[i+j],v=p[i+l+j];
6         if(!inv)p[i+j]=u+v,p[i+l+j]=u-v; // XOR
7         else p[i+j]=(u+v)/2,p[i+l+j]=(u-v)/2;
8         //if(!inv)p[i+j]=v,p[i+l+j]=u+v; // AND
9         //else p[i+j]=-u+v,p[i+l+j]=u;
10        //if(!inv)p[i+j]=u+v,p[i+l+j]=u; // OR
11        //else p[i+j]=v,p[i+l+j]=u-v;
12    }
13    // like polynomial multiplication, but XORing exponents
14    // instead of adding them (also ANDing, ORing)
15    vector<ll> multiply(vector<ll>& p1, vector<ll>& p2){
16        int n=1<<(32-__builtin_clz(max(SZ(p1),SZ(p2))-1));
17        for (int i = 0; i < n; ++i)c1[i]=0,c2[i]=0;
18        for (int i = 0; i < SZ(p1); ++i)c1[i]=p1[i];
19        for (int i = 0; i < SZ(p2); ++i)c2[i]=p2[i];
20        fht(c1,n,false);fht(c2,n,false);
21        for (int i = 0; i < n; ++i)c1[i]*=c2[i];
22        fht(c1,n,true);
23        return vector<ll>(c1,c1+n);
24    }

```

4.13 Karatsuba

```

1 typedef ll tp;
2 #define add(n,s,d,k) for (int i = 0; i < n; ++i)(d)[i]+=(s)[i]*k
3 tp* ini(int n){tp *r=new tp[n];fill(r,r+n,0);return r;}

```

```

4 void karatsura(int n, tp* p, tp* q, tp* r){
5     if(n<=0)return;
6     if(n<35)for (int i = 0; i < n; ++i)for (int j = 0; j < n; ++j)r[i+j]+=
7         p[i]*q[j];
8     else {
9         int nac=n/2,nbd=n-n/2;
10        tp *a=p,*b=p+nac,*c=q,*d=q+nac;
11        tp *ab=ini(nbd+1),*cd=ini(nbd+1),*ac=ini(nac*2),*bd=ini(nbd*2);
12        add(nac,a,ab,1);add(nbd,b,ab,1);
13        add(nac,c,cd,1);add(nbd,d,cd,1);
14        karatsura(nac,a,c,ac);karatsura(nbd,b,d,bd);
15        add(nac*2,ac,r+nac,-1);add(nbd*2,bd,r+nac,-1);
16        add(nac*2,ac,r,1);add(nbd*2,bd,r+nac*2,1);
17        karatsura(nbd+1,ab,cd,r+nac);
18        free(ab);free(cd);free(ac);free(bd);
19    }
20    vector<tp> multiply(vector<tp> p0, vector<tp> p1){
21        int n=max(p0.size(),p1.size());
22        tp *p=ini(n),*q=ini(n),*r=ini(2*n);
23        for (int i = 0; i < p0.size(); ++i)p[i]=p0[i];
24        for (int i = 0; i < p1.size(); ++i)q[i]=p1[i];
25        karatsura(n,p,q,r);
26        vector<tp> rr(r,r+p0.size()+p1.size()-1);
27        free(p);free(q);free(r);
28        return rr;
29    }

```

4.14 Diophantine

```

1 pair<ll,ll> extendedEuclid (ll a, ll b){ //a * x + b * y = gcd(a,b)
2     ll x,y;
3     if (b==0) return {1,0};
4     auto p=extendedEuclid(b,a%b);
5     x=p.second;
6     y=p.first-(a/b)*x;
7     if(a*x+b*y==gcd(a,b)) x=-x, y=-y;
8     return {x,y};
9 }
10 pair<pair<ll,ll>,pair<ll,ll> > diophantine(ll a,ll b, ll r) {
11     //a*x+b*y=r where r is multiple of gcd(a,b);
12     ll d=gcd(a,b);
13     a/=d; b/=d; r/=d;

```

```

14 auto p = extendedEuclid(a,b);
15 p.first*=r; p.second*=r;
16 assert(a*p.first+b*p.second==r);
17 return {p,{-b,a}}; // solutions: p+t*ans.second
18 }

```

4.15 Modular inverse

```

1 ll inv(ll a, ll mod) { //inverse of a modulo mod
2   assert(gcd(a,mod)==1);
3   pl sol = extendedEuclid(a,mod);
4   return ((sol.first%mod)+mod)%mod;
5 }

```

4.16 Chinese remainder theorem

```

1 #define mod(a,m) (((a)%m+m)%m)
2 pair<ll,ll> sol(tuple<ll,ll,ll> c){ //requires inv, diophantine
3   ll a=get<0>(c), x1=get<1>(c), m=get<2>(c), d=gcd(a,m);
4   if(d==1) return {mod(x1*inv(a,m),m), m};
5   else return x1%d ? ii({-1LL,-1LL}) : sol(make_tuple(a/d,x1/d,m/d));
6 }
7 pair<ll,ll> crt(vector< tuple<ll,ll,ll> > cond) { // returns: (sol, lcm)
8   ll x1=0,m1=1,x2,m2;
9   for(auto t:cond){
10    tie(x2,m2)=sol(t);
11    if(m2<0|| (x1-x2)%gcd(m1,m2))return {-1,-1};
12    if(m1==m2)continue;
13    ll k=diophantine(m2,-m1,x1-x2).first.second,l=m1*(m2/gcd(m1,m2));
14    x1=mod((__int128)m1*k+x1,l);m1=l;
15  }
16  return sol(make_tuple(1,x1,m1));
17 } //cond[i]={ai,bi,mi} ai*xi=bi (mi); assumes lcm fits in ll

```

4.17 Mobius

```

1 short mu[MAXN] = {0,1};
2 void mobius(){
3   for (int i = 1; i < MAXN; ++i)if(mu[i])for(int j=i+i;j<MAXN;j+=i)mu[j]
4     -=mu[i];
5 }

```

4.18 Matrix exponentiation

```

1 typedef vector<vector<ll> > Matrix;
2 Matrix ones(int n) {
3   Matrix r(n,vector<ll>(n));
4   for (int i = 0; i < n; ++i)r[i][i]=1;
5   return r;
6 }
7 Matrix operator*(Matrix &a, Matrix &b) {
8   int n=SZ(a),m=SZ(b[0]),z=SZ(a[0]);
9   Matrix r(n,vector<ll>(m));
10  for (int i = 0; i < n; ++i)for (int j = 0; j < m; ++j)for (int k = 0;
11    k < z; ++k)
12    r[i][j]+=a[i][k]*b[k][j],r[i][j]%mod;
13  return r;
14 }
15 Matrix be(Matrix b, ll e) {
16   Matrix r=ones(SZ(b));
17   while(e){if(e&1LL)r=r*b;b=b*b;e/=2;}
18   return r;
19 }

```

4.19 Matrix reduce and determinant

```

1 double reduce(vector<vector<double> >& x){ // returns determinant
2   int n=x.size(),m=x[0].size();
3   int i=0,j=0;double r=1.;
4   while(i<n&&j<m){
5     int l=i;
6     for (int k = i+1; k < n; ++k)if(abs(x[k][j])>abs(x[l][j]))l=k;
7     if(abs(x[l][j])<EPS){j++;r=0.;continue;}
8     if(l!=i){r=-r;swap(x[i],x[l]);}
9     r*=x[i][j];
10    for(int k=m-1;k>=j;k--)x[i][k]/=x[i][j];
11    for (int k = 0; k < n; ++k){
12      if(k==i)continue;
13      for(int l=m-1;l>=j;l--)x[k][l]-=x[k][j]*x[i][l];
14    }
15    i++;j++;
16  }
17  return r;
18 }

```

4.20 Simplex

```

1 vector<int> X,Y;

```

```

2 vector<vector<double> > A;
3 vector<double> b,c;
4 double z;
5 int n,m;
6 void pivot(int x,int y){
7     swap(X[y],Y[x]);
8     b[x]/=A[x][y];
9     for (int i = 0; i < m; ++i)if(i!=y)A[x][i]/=A[x][y];
10    A[x][y]=1/A[x][y];
11    for (int i = 0; i < n; ++i)if(i!=x&&abs(A[i][y])>EPS){
12        b[i]-=A[i][y]*b[x];
13        for (int j = 0; j < m; ++j)if(j!=y)A[i][j]-=A[i][y]*A[x][j];
14        A[i][y]=-A[i][y]*A[x][y];
15    }
16    z+=c[y]*b[x];
17    for (int i = 0; i < m; ++i)if(i!=y)c[i]-=c[y]*A[x][i];
18    c[y]=-c[y]*A[x][y];
19 }
20 pair<double,vector<double> > simplex( // maximize c^T x s.t. Ax<=b, x>=0
21     vector<vector<double> > _A, vector<double> _b, vector<double> _c){
22     // returns pair (maximum value, solution vector)
23     A=_A;b=_b;c=_c;
24     n=b.size();m=c.size();z=0.;
25     X=vector<int>(m);Y=vector<int>(n);
26     for (int i = 0; i < m; ++i)X[i]=i;
27     for (int i = 0; i < n; ++i)Y[i]=i+m;
28     while(1){
29         int x=-1,y=-1;
30         double mn=-EPS;
31         for (int i = 0; i < n; ++i)if(b[i]<mn)mn=b[i],x=i;
32         if(x<0)break;
33         for (int i = 0; i < m; ++i)if(A[x][i]<-EPS){y=i;break;}
34         assert(y>0); // no solution to Ax<=b
35         pivot(x,y);
36     }
37     while(1){
38         double mx=EPS;
39         int x=-1,y=-1;
40         for (int i = 0; i < m; ++i)if(c[i]>mx)mx=c[i],y=i;
41         if(y<0)break;
42         double mn=1e200;
43         for (int i = 0; i < n; ++i)if(A[i][y]>EPS&&b[i]/A[i][y]<mn)mn=b[i]/A
            [i][y],x=i;

```

```

44     assert(x>0); // c^T x is unbounded
45     pivot(x,y);
46 }
47 vector<double> r(m);
48 for (int i = 0; i < n; ++i)if(Y[i]<m)r[Y[i]]=b[i];
49 return {z,r};
50 }

```

4.21 Discrete log

```

1 //returns x such that a^x = b (mod m) or -1 if inexistent
2 ll discrete_log(ll a,ll b,ll m) {
3     a%=m, b%=m;
4     if(b == 1) return 0;
5     int cnt=0;
6     ll tmp=1;
7     for(int g=__gcd(a,m);g!=1;g=__gcd(a,m)) {
8         if(b%g) return -1;
9         m/=g, b/=g;
10        tmp = tmp*a/g%m;
11        ++cnt;
12        if(b == tmp) return cnt;
13    }
14    map<ll,int> w;
15    int s = ceil(sqrt(m));
16    ll base = b;
17    for (int i = 0; i < s; ++i) {
18        w[base] = i;
19        base=base*a%m;
20    }
21    base=fastpow(a,s,m);
22    ll key=tmp;
23    for (int i = 1; i < s+2; ++i) {
24        key=base*key%m;
25        if(w.count(key)) return i*s-w[key]+cnt;
26    }
27    return -1;
28 }

```

4.22 Berlekamp Massey

```

1 typedef vector<int> vi;
2 vi BM(vi x){
3     vi ls,cur;int lf,ld;

```



```

4   for (int i = 0; i < SZ(x); ++i){
5       ll t=0;
6       for (int j = 0; j < SZ(cur); ++j)t=(t+x[i-j-1]*(ll)cur[j])%MOD;
7       if((t-x[i])%MOD==0)continue;
8       if(!SZ(cur)){cur.resize(i+1);lf=i;ld=(t-x[i])%MOD;continue;}
9       ll k=-(x[i]-t)*fast_pow(ld,MOD-2)%MOD;
10      vi c(i-lf-1);c.push_back(k);
11      for (int j = 0; j < SZ(ls); ++j)c.push_back((-ls[j]*k)%MOD);
12      if(SZ(c)<SZ(cur))c.resize(SZ(cur));
13      for (int j = 0; j < SZ(cur); ++j)c[j]=(c[j]+cur[j])%MOD;
14      if(i-lf+SZ(ls)>=SZ(cur))ls=cur,lf=i,ld=(t-x[i])%MOD;
15      cur=c;
16  }
17  for (int i = 0; i < SZ(cur); ++i)cur[i]=(cur[i]%MOD+MOD)%MOD;
18  return cur;
19 }

```

4.23 Linear Rec

```

1  //init O(n^2log) query(n^2 logk)
2  //input: terms: first n term; trans: transition function; MOD; LOG=mxlog
   of k
3  //output calc(k): kth term mod MOD
4  //example: {1,1} {2,1} an=2*a(n-1)+a(n-2); calc(3)=3 calc(10007)
   =71480733
5  struct LinearRec{
6      typedef vector<int> vi;
7      int n; vi terms, trans; vector<vi> bin;
8      vi add(vi &a, vi &b){
9          vi res(n*2+1);
10         for (int i = 0; i < n+1; ++i)for (int j = 0; j < n+1; ++j)res[i+j]=
            res[i+j]*1LL+(ll)a[i]*b[j])%MOD;
11         for(int i=2*n; i>n; --i){
12             for (int j = 0; j < n; ++j)res[i-1-j]=(res[i-1-j]*1LL+(ll)res[i]*
                trans[j])%MOD;
13             res[i]=0;
14         }
15         res.erase(res.begin()+n+1,res.end());
16         return res;
17     }
18     LinearRec(vi &terms, vi &trans):terms(terms),trans(trans){
19         n=SZ(trans);vi a(n+1);a[1]=1;
20         bin.push_back(a);

```

```

21     for (int i = 1; i < LOG; ++i)bin.push_back(add(bin[i-1],bin[i-1]));
22 }
23 int calc(int k){
24     vi a(n+1);a[0]=1;
25     for (int i = 0; i < LOG; ++i)if((k>>i)&1)a=add(a,bin[i]);
26     int ret=0;
27     for (int i = 0; i < n; ++i)ret=((ll)ret+(ll)a[i+1]*terms[i])%MOD;
28     return ret;
29 }
30 };

```

4.24 Tonelli Shanks

```

1  ll legendre(ll a, ll p){
2      if(a%p==0)return 0; if(p==2)return 1;
3      return fpow(a,(p-1)/2,p);
4  }
5  ll tonelli_shanks(ll n, ll p){ // sqrt(n) mod p (p must be a prime)
6      assert(legendre(n,p)==1); if(p==2)return 1;
7      ll s=__builtin_ctzll(p-1), q=(p-1LL)>>s, z=rnd(1,p-1);
8      if(s==1)return fpow(n,(p+1)/4LL,p);
9      while(legendre(z,p)!=p-1)z=rnd(1,p-1);
10     ll c=fpow(z,q,p), r=fpow(n,(q+1)/2,p), t=fpow(n,q,p), m=s;
11     while(t!=1){
12         ll i=1, ts=(t*t)%p;
13         while(ts!=1)i++,ts=(ts*ts)%p;
14         ll b=c;
15         for (int _ = 0; _ < m-i-1; ++_)b=(b*b)%p;
16         r=r*b%p;c=b*b%p;t=t*c%p;m=i;
17     }
18     return r;
19 }

```

4.25 Prime count

```

1  // Count number of primes in [1,n]
2  // O(n^(2/3)) time
3  // O(sqrt(n)) memory
4  ll prime_count(ll n) {
5      if (n <= 1) return 0;
6      int v = sqrtl(n), s = (v+1)>>1, pc=0;
7      vector<int> smalls(s), roughs(s), skip(v+1);
8      vector<ll> larges(s);
9      for (int i = 0; i < s; ++i){

```



```

10     smalls[i]=i;
11     roughs[i]=2*i+1;
12     larges[i]=(n/roughs[i]-1)>>1;
13 }
14 for(int p=3;p<=v;p+=2) if(!skip[p]){
15     int q=p*p;
16     if(1ll*q*q>n) break;
17     skip[p]=1;
18     for(int i=q;i<=v;i+=2*p) skip[i]=1;
19     int ns=0;
20     for (int k = 0; k < s; ++k){
21         int i=roughs[k];
22         if(skip[i]) continue;
23         ll d=1ll*i*p;
24         larges[ns]=larges[k]-(d<=v?larges[smalls[d]>>1]-pc]:smalls[(n
            /d-1)>>1])+pc;
25         roughs[ns++]=i;
26     }
27     s=ns;
28     for(int i=(v-1)>>1, j=((v/p)-1)|1; j>=p; j-=2){
29         int c=smalls[j]>>1]-pc;
30         for(int e=(j*p)>>1; i>=e; i--) smalls[i]-=c;
31     }
32     pc++;
33 }
34 larges[0]+=1ll*(s+2*(pc-1))*(s-1)>>1;
35 for (int k = 1; k < s; ++k) larges[0]-=larges[k];
36 for (int l = 1; l < s; ++l){
37     int q=roughs[l];
38     ll m=n/q, t=0;
39     int e=smalls[(m/q-1)>>1]-pc;
40     if(e<1+1)break;
41     for (int k = l+1; k < e+1; ++k) t+=smalls[(m/roughs[k]-1)>>1];
42     larges[0]+=t-1ll*(e-l)*(pc+l-1);
43 }
44 return larges[0]+1;
45 }

```

5 Geometry

5.1 Point

```

1 struct pt { // for 3D add z coordinate

```

```

2     double x,y;
3     pt(double x, double y):x(x),y(y){}
4     pt(){}
5     double norm2(){return *this**this;}
6     double norm(){return sqrt(norm2());}
7     bool operator==(pt p){return abs(x-p.x)<=EPS&&abs(y-p.y)<=EPS;}
8     pt operator+(pt p){return pt(x+p.x,y+p.y);}
9     pt operator-(pt p){return pt(x-p.x,y-p.y);}
10    pt operator*(double t){return pt(x*t,y*t);}
11    pt operator/(double t){return pt(x/t,y/t);}
12    double operator*(pt p){return x*p.x+y*p.y;}
13    // pt operator^(pt p){ // only for 3D
14    //     return pt(y*p.z-z*p.y,z*p.x-x*p.z,x*p.y-y*p.x);}
15    double angle(pt p){ // redefine acos for values out of range
16        return acos(*this*p/(norm()*p.norm()));}
17    pt unit(){return *this/norm();}
18    double operator%(pt p){return x*p.y-y*p.x;}
19    // 2D from now on
20    bool operator<(pt p)const{ // for convex hull
21        return x<p.x-EPS|| (abs(x-p.x)<=EPS&&y<p.y-EPS);}
22    bool left(pt p, pt q){ // is it to the left of directed line pq?
23        return (q-p)%(*this-p)>EPS;}
24    pt rot(pt r){return pt(*this%r,*this*r);}
25    pt rot(double a){return rot(pt(sin(a),cos(a)));}
26 };
27 pt ccw90(1,0);
28 pt cw90(-1,0);

```

5.2 Line

```

1 int sgn2(double x){return x<0?-1:1;}
2 struct ln {
3     pt p,pq;
4     ln(pt p, pt q):p(p),pq(q-p){}
5     ln(){}
6     bool has(pt r){return dist(r)<=EPS;}
7     bool seghas(pt r){return has(r)&&(r-p)*(r-(p+pq))<=EPS;}
8     // bool operator /(ln l){return (pq.unit()^l.pq.unit()).norm()<=EPS;}
9     // 3D
10    bool operator/(ln l){return abs(pq.unit()%l.pq.unit())<=EPS;} // 2D
11    bool operator==(ln l){return *this/l&&has(l.p);}
12    pt operator^(ln l){ // intersection
13        if(*this/l)return pt(DINF,DINF);

```

```

13     pt r=l.p+l.pq*((p-l.p)%pq/(l.pq%pq));
14     // if(!has(r)){return pt(NAN,NAN,NAN);} // check only for 3D
15     return r;
16 }
17 double angle(ln l){return pq.angle(l.pq);}
18 int side(pt r){return has(r)?0:sgn2(pq%(r-p));} // 2D
19 pt proj(pt r){return p+pq*((r-p)*pq/pq.norm2());}
20 pt ref(pt r){return proj(r)*2-r;}
21 double dist(pt r){return (r-proj(r)).norm();}
22 // double dist(ln l){ // only 3D
23 //     if(*this/l)return dist(l.p);
24 //     return abs((l.p-p)*(pq^l.pq))/(pq^l.pq).norm());
25 // }
26 ln rot(auto a){return ln(p,p+pq.rot(a));} // 2D
27 };
28 ln bisector(ln l, ln m){ // angle bisector
29     pt p=l^m;
30     return ln(p,p+l.pq.unit()+m.pq.unit());
31 }
32 ln bisector(pt p, pt q){ // segment bisector (2D)
33     return ln((p+q)*.5,p).rot(ccw90);
34 }

```

5.3 Circle

```

1 struct circle {
2     pt o;double r;
3     circle(pt o, double r):o(o),r(r){}
4     circle(pt x, pt y, pt z){o=bisector(x,y)^bisector(x,z);r=(o-x).norm()
5         ;}
6     bool has(pt p){return (o-p).norm()<=r+EPS;}
7     vector<pt> operator^(circle c){ // ccw
8         vector<pt> s;
9         double d=(o-c.o).norm();
10        if(d>r+c.r+EPS||d+min(r,c.r)+EPS<max(r,c.r))return s;
11        double x=(d*d-c.r*c.r+r*r)/(2*d);
12        double y=sqrt(r*r-x*x);
13        pt v=(c.o-o)/d;
14        s.push_back(o+v*x-v.rot(ccw90)*y);
15        if(y>EPS)s.push_back(o+v*x+v.rot(ccw90)*y);
16        return s;
17    }
18    vector<pt> operator^(ln l){

```

```

18    vector<pt> s;
19    pt p=l.proj(o);
20    double d=(p-o).norm();
21    if(d-EPS>r)return s;
22    if(abs(d-r)<=EPS){s.push_back(p);return s;}
23    d=sqrt(r*r-d*d);
24    s.push_back(p+l.pq.unit()*d);
25    s.push_back(p-l.pq.unit()*d);
26    return s;
27 }
28 vector<pt> tang(pt p){
29     double d=sqrt((p-o).norm2()-r*r);
30     return *this^circle(p,d);
31 }
32 bool in(circle c){ // non strict
33     double d=(o-c.o).norm();
34     return d+r<=c.r+EPS;
35 }
36 double intertriangle(pt a, pt b){ // area of intersection with oab
37     if(abs((o-a)%(o-b))<=EPS)return 0.;
38     vector<pt> q={a},w=*this^ln(a,b);
39     if(w.size()==2)for(auto p:w)if((a-p)*(b-p)<-EPS)q.push_back(p);
40     q.push_back(b);
41     if(q.size()==4&&(q[0]-q[1])*(q[2]-q[1])>EPS)swap(q[1],q[2]);
42     double s=0;
43     for (int i = 0; i < q.size()-1; ++i){
44         if(!has(q[i])||!has(q[i+1]))s+=r*r*(q[i]-o).angle(q[i+1]-o)/2;
45         else s+=abs((q[i]-o)%(q[i+1]-o)/2);
46     }
47     return s;
48 }
49 };
50 vector<double> intercircles(vector<circle> c){
51     vector<double> r(SZ(c)+1); // r[k]: area covered by at least k circles
52     for (int i = 0; i < SZ(c); ++i){ // 0(n^2 log n) (high
53         constant)
54         int k=1;Cmp s(c[i].o);
55         vector<pair<pt,int> > p={
56             {c[i].o+pt(1,0)*c[i].r,0},
57             {c[i].o-pt(1,0)*c[i].r,0}};
58         for (int j = 0; j < SZ(c); ++j)if(j!=i){
59             bool b0=c[i].in(c[j]),b1=c[j].in(c[i]);
60             if(b0&&(!b1||i<j))k++;

```

```

60     else if(!b0&&!b1){
61         auto v=c[i]^c[j];
62         if(SZ(v)==2){
63             p.push_back({v[0],1});p.push_back({v[1],-1});
64             if(s(v[1],v[0]))k++;
65         }
66     }
67 }
68 sort(p.begin(),p.end(),
69 [&](pair<pt,int> a, pair<pt,int> b){return s(a.first,b.first);});
70 for (int j = 0; j < SZ(p); ++j){
71     pt p0=p[j-1:SZ(p)-1].first,p1=p[j].first;
72     double a=(p0-c[i].o).angle(p1-c[i].o);
73     r[k]+=(p0.x-p1.x)*(p0.y+p1.y)/2+c[i].r*c[i].r*(a-sin(a))/2;
74     k+=p[j].second;
75 }
76 }
77 return r;
78 }

```

5.4 Polygon

```

1 int sgn(double x){return x<-EPS?-1:x>EPS;}
2 struct pol {
3     int n;vector<pt> p;
4     pol(){}
5     pol(vector<pt> _p){p=_p;n=p.size();}
6     double area(){
7         double r=0.;
8         for (int i = 0; i < n; ++i)r+=p[i]%p[(i+1)%n];
9         return abs(r)/2; // negative if CW, positive if CCW
10    }
11    pt centroid(){ // (barycenter)
12        pt r(0,0);double t=0;
13        for (int i = 0; i < n; ++i){
14            r=r+(p[i]+p[(i+1)%n])*p[i]%p[(i+1)%n];
15            t+=p[i]%p[(i+1)%n];
16        }
17        return r/t/3;
18    }
19    bool has(pt q){ // O(n)
20        for (int i = 0; i < n; ++i)if(ln(p[i],p[(i+1)%n]).seghas(q))return
            true;

```

```

21    int cnt=0;
22    for (int i = 0; i < n; ++i){
23        int j=(i+1)%n;
24        int k=sgn((q-p[j])%(p[i]-p[j]));
25        int u=sgn(p[i].y-q.y),v=sgn(p[j].y-q.y);
26        if(k>0&&u<0&&v>=0)cnt++;
27        if(k<0&&v<0&&u>=0)cnt--;
28    }
29    return cnt!=0;
30 }
31 void normalize(){ // (call before haslog, remove collinear first)
32     if(p[2].left(p[0],p[1]))reverse(p.begin(),p.end());
33     int pi=min_element(p.begin(),p.end())-p.begin();
34     vector<pt> s(n);
35     for (int i = 0; i < n; ++i)s[i]=p[(pi+i)%n];
36     p.swap(s);
37 }
38 bool haslog(pt q){ // O(log(n)) only CONVEX. Call normalize first
39     if(q.left(p[0],p[1])||q.left(p.back(),p[0]))return false;
40     int a=1,b=p.size()-1; // returns true if point on boundary
41     while(b-a>1){ // (change sign of EPS in left
42         int c=(a+b)/2; // to return false in such case)
43         if(!q.left(p[0],p[c]))a=c;
44         else b=c;
45     }
46     return !q.left(p[a],p[a+1]);
47 }
48 pt farthest(pt v){ // O(log(n)) only CONVEX
49     if(n<10){
50         int k=0;
51         for (int i = 1; i < n; ++i)if(v*(p[i]-p[k])>EPS)k=i;
52         return p[k];
53     }
54     if(n==SZ(p))p.push_back(p[0]);
55     pt a=p[1]-p[0];
56     int s=0,e=n,ua=v*a>EPS;
57     if(!ua&&v*(p[n-1]-p[0])<=EPS)return p[0];
58     while(1){
59         int m=(s+e)/2;pt c=p[m+1]-p[m];
60         int uc=v*c>EPS;
61         if(!uc&&v*(p[m-1]-p[m])<=EPS)return p[m];
62         if(ua&&(!uc|v*(p[s]-p[m])>EPS))e=m;
63         else if(ua|uc|v*(p[s]-p[m])>=-EPS)s=m,a=c,ua=uc;

```

```

64     else e=m;
65     assert(e>s+1);
66 }
67 }
68 pol cut(ln l){ // cut CONVEX polygon by line l
69     vector<pt> q; // returns part at left of l.pq
70     for (int i = 0; i < n; ++i){
71         int d0=sgn(l.pq%(p[i]-l.p)),d1=sgn(l.pq%(p[(i+1)%n]-l.p));
72         if(d0>=0)q.push_back(p[i]);
73         ln m(p[i],p[(i+1)%n]);
74         if(d0*d1<0&&!(l/m)q.push_back(l~m);
75     }
76     return pol(q);
77 }
78 double intercircle(circle c){ // area of intersection with circle
79     double r=0.;
80     for (int i = 0; i < n; ++i){
81         int j=(i+1)%n;double w=c.intertriangle(p[i],p[j]);
82         if((p[j]-c.o)%(p[i]-c.o)>0)r+=w;
83         else r-=w;
84     }
85     return abs(r);
86 }
87 double callipers(){ // square distance of most distant points
88     double r=0; // prereq: convex, ccw, NO COLLINEAR POINTS
89     for(int i=0,j=n<2?0:1;i<j;++i){
90         for(;;j=(j+1)%n){
91             r=max(r,(p[i]-p[j]).norm2());
92             if((p[(i+1)%n]-p[i])%(p[(j+1)%n]-p[j])<=EPS)break;
93         }
94     }
95     return r;
96 }
97 };
98 // Dynamic convex hull trick
99 vector<pol> w;
100 void add(pt q){ // add(q), O(log^2(n))
101     vector<pt> p={q};
102     while(!w.empty()&&SZ(w.back().p)<2*SZ(p)){
103         for(pt v:w.back().p)p.push_back(v);
104         w.pop_back();
105     }
106     w.push_back(pol(chull(p)));

```

```

107 }
108 ll query(pt v){ // max(q*v:q in w), O(log^2(n))
109     ll r=-INF;
110     for(auto& p:w)r=max(r,p.farthest(v)*v);
111     return r;
112 }

```

5.5 Plane

```

1 struct plane {
2     pt a,n; // n: normal unit vector
3     plane(pt a, pt b, pt c):a(a),n(((b-a)^(c-a)).unit()){}
4     plane(){}
5     bool has(pt p){return abs((p-a)*n)<=EPS;}
6     double angle(plane w){return acos(n*w.n);}
7     double dist(pt p){return abs((p-a)*n);}
8     pt proj(pt p){inter(ln(p,p+n),p);return p;}
9     bool inter(ln l, pt& r){
10         double x=n*(l.p+l.pq-a),y=n*(l.p-a);
11         if(abs(x-y)<=EPS)return false;
12         r=(l.p*x-(l.p+l.pq)*y)/(x-y);
13         return true;
14     }
15     bool inter(plane w, ln& r){
16         pt nn=n^w.n;pt v=n^nn;double d=w.n*v;
17         if(abs(d)<=EPS)return false;
18         pt p=a+v*(w.n*(w.a-a)/d);
19         r=ln(p,p+nn);
20         return true;
21     }
22 };

```

5.6 Radial order of points

```

1 struct Cmp { // IMPORTANT: add const in pt operator -
2     pt r;
3     Cmp(pt r):r(r){}
4     int cuad(const pt &a)const {
5         if(a.x>0&&a.y>=0)return 0;
6         if(a.x<=0&&a.y>0)return 1;
7         if(a.x<0&&a.y<=0)return 2;
8         if(a.x>=0&&a.y<0)return 3;
9         assert(a.x==0&&a.y==0);
10        return -1;

```

```

11 }
12 bool cmp(const pt& p1, const pt& p2) const {
13     int c1=cuad(p1),c2=cuad(p2);
14     if(c1==c2) return p1.y*p2.x<p1.x*p2.y;
15     return c1<c2;
16 }
17 bool operator()(const pt& p1, const pt& p2) const {
18     return cmp(p1-r,p2-r);
19 }
20 };

```

5.7 Convex hull

```

1 // CCW order
2 // Includes collinear points (change sign of EPS in left to exclude)
3 vector<pt> chull(vector<pt> p){
4     vector<pt> r;
5     sort(p.begin(),p.end()); // first x, then y
6     p.erase(unique(ALL(p)), p.end());
7     for (int i = 0; i < p.size(); ++i){ // lower hull
8         while(r.size()>=2&&r.back().left(r[r.size()-2],p[i]))r.pop_back();
9         r.push_back(p[i]);
10    }
11    r.pop_back();
12    int k=r.size(),pr=k-1;
13    for(int i=p.size()-1;i>=0;--i){ // upper hull
14        while(r.size()>=k+2&&r.back().left(r[r.size()-2],p[i]))r.pop_back();
15        while(pr>=0&&p[i]<r[pr]) pr--;
16        if(pr<0||!(p[i]==r[pr])) r.push_back(p[i]);
17    }
18    return r;
19 }

```

5.8 Dual from planar graph

```

1 vector<int> g[MAXN];int n; // input graph (must be connected)
2 vector<int> gd[MAXN];int nd; // output graph
3 vector<int> nodes[MAXN]; // nodes delimiting region (in CW order)
4 map<pair<int,int>,int> ps,es;
5 void get_dual(vector<pt> p){ // p: points corresponding to nodes
6     ps.clear();es.clear();
7     for (int x = 0; x < n; ++x){
8         Cmp pc(p[x]); // (radial order of points)
9         auto comp=[&](int a, int b){return pc(p[a],p[b]);};

```

```

10     sort(g[x].begin(),g[x].end(),comp);
11     for (int i = 0; i < g[x].size(); ++i)ps[{x,g[x][i]}]=i;
12 }
13 nd=0;
14 for (int xx = 0; xx < n; ++xx)for(auto yy:g[xx])if(!es.count({xx,yy}))
15 {
16     int x=xx,y=yy;gd[nd].clear();nodes[nd].clear();
17     while(!es.count({x,y})){
18         es[{x,y}]=nd;nodes[nd].push_back(y);
19         int z=g[y][ (ps[{y,x}]+1)%g[y].size() ];x=y;y=z;
20     }
21     nd++;
22 }
23 for(auto p:es){
24     pair<int,int> q={p.first.second,p.first.first};
25     assert(es.count(q));
26     if(es[q]!=p.second)gd[p.second].push_back(es[q]);
27 }
28 for (int i = 0; i < nd; ++i){
29     sort(gd[i].begin(),gd[i].end());
30     gd[i].erase(unique(gd[i].begin(),gd[i].end()),gd[i].end());
31 }

```

5.9 Halfplane intersection

```

1 // polygon intersecting left side of halfplanes
2 struct halfplane:public ln{
3     double angle;
4     halfplane(){}
5     halfplane(pt a,pt b){p=a; pq=b-a; angle=atan2(pq.y,pq.x);}
6     bool operator<(halfplane b)const{return angle<b.angle;}
7     bool out(pt q){return pq%(q-p)<=-EPS;}
8 };
9 vector<pt> intersect(vector<halfplane> b){
10     vector<pt> bx={{DINF,DINF},{-DINF,DINF},{-DINF,-DINF},{DINF,-DINF}};
11     for (int i = 0; i < 4; ++i) b.push_back(halfplane(bx[i],bx[(i+1)%4]));
12     sort(ALL(b));
13     int n=SZ(b),q=1,h=0;
14     vector<halfplane> c(SZ(b)+10);
15     for (int i = 0; i < n; ++i){
16         while(q<h&&b[i].out(c[h]^c[h-1])) h--;
17         while(q<h&&b[i].out(c[q]^c[q+1])) q++;

```

```

18     c[++h]=b[i];
19     if(q<h&&abs(c[h].pq%c[h-1].pq)<EPS){
20         if(c[h].pq*c[h-1].pq<=0) return {};
21         h--;
22         if(b[i].out(c[h].p)) c[h]=b[i];
23     }
24 }
25 while(q<h-1&&c[q].out(c[h]^c[h-1]))h--;
26 while(q<h-1&&c[h].out(c[q]^c[q+1]))q++;
27 if(h-q<=1)return {};
28 c[h+1]=c[q];
29 vector<pt> s;
30 for (int i = q; i < h+1; ++i) s.push_back(c[i]^c[i+1]);
31 return s;
32 }

```

5.10 KD tree

```

1 // given a set of points, answer queries of nearest point in O(log(n))
2 bool onx(pt a, pt b){return a.x<b.x;}
3 bool ony(pt a, pt b){return a.y<b.y;}
4 struct Node {
5     pt pp;
6     ll x0=INF, x1=-INF, y0=INF, y1=-INF;
7     Node *first=0, *second=0;
8     ll distance(pt p){
9         ll x=min(max(x0,p.x),x1);
10        ll y=min(max(y0,p.y),y1);
11        return (pt(x,y)-p).norm2();
12    }
13    Node(vector<pt>&& vp):pp(vp[0]){
14        for(pt p:vp){
15            x0=min(x0,p.x); x1=max(x1,p.x);
16            y0=min(y0,p.y); y1=max(y1,p.y);
17        }
18        if(SZ(vp)>1){
19            sort(ALL(vp),x1-x0>=y1-y0?onx:ony);
20            int m=SZ(vp)/2;
21            first=new Node({vp.begin(),vp.begin()+m});
22            second=new Node({vp.begin()+m,vp.end()});
23        }
24    }
25 };

```

```

26 struct KDTree {
27     Node* root;
28     KDTree(const vector<pt>& vp):root(new Node({ALL(vp)})) {}
29     pair<ll,pt> search(pt p, Node *node){
30         if(!node->first){
31             //avoid query point as answer
32             //if(p==node->pp) {INF,pt()};
33             return {(p-node->pp).norm2(),node->pp};
34         }
35         Node *f=node->first, *s=node->second;
36         ll bf=f->distance(p), bs=s->distance(p);
37         if(bf>bs)swap(bf,bs),swap(f,s);
38         auto best=search(p,f);
39         if(bs<best.first) best=min(best,search(p,s));
40         return best;
41     }
42     pair<ll,pt> nearest(pt p){return search(p,root);}
43 };

```

5.11 Minkowski sum

```

1 // Compute Minkowski sum of two CONVEX polygons (remove collinear first)
2 // Returns answer in CCW order
3 void reorder(vector<pt> &p){
4     if(!p[2].left(p[0],p[1])) reverse(ALL(p));
5     int pos=0;
6     for (int i = 1; i < SZ(p); ++i) if(pt(p[i].y,p[i].x) < pt(p[pos].y,p[
7         pos].x)) pos=i;
8     rotate(p.begin(), p.begin()+pos, p.end());
9 }
10 vector<pt> minkowski_sum(vector<pt> p, vector<pt> q){
11     if(min(SZ(p),SZ(q))<3){
12         vector<pt> v;
13         for(pt pp:p) for(pt qq:q) v.push_back(pp+qq);
14         return hull(v);
15     }
16     reorder(p); reorder(q);
17     for (int i = 0; i < 2; ++i) p.push_back(p[i]), q.push_back(q[i]);
18     vector<pt> r;
19     int i=0,j=0;
20     while(i+2<SZ(p)||j+2<SZ(q)){
21         r.push_back(p[i]+q[j]);
22         auto cross=(p[i+1]-p[i])%(q[j+1]-q[j]);

```

```

22     i+=cross>=-EPS;
23     j+=cross<=EPS;
24 }
25 return r;
26 }

```

5.12 Delaunay triangulation

```

1 // Returns planar graph representing Delaunay's triangulation.
2 // Edges for each vertex are in ccw order.
3 // It can work with doubles, but also integers (replace long double in
4 // line 51)
5 typedef struct QuadEdge* Q;
6 struct QuadEdge {
7     int id,used;
8     pt o;
9     Q rot,nxt;
10    QuadEdge(int id=-1, pt o=pt(INF,INF)):id(id),used(0),o(o),rot(0),nxt
11    (0){}
12    Q rev(){return rot->rot;}
13    Q next(){return nxt;}
14    Q prev(){return rot->next()->rot;}
15    pt dest(){return rev()->o;}
16 };
17
18 Q edge(pt a, pt b, int ida, int idb){
19     Q e1=new QuadEdge(ida,a);
20     Q e2=new QuadEdge(idb,b);
21     Q e3=new QuadEdge;
22     Q e4=new QuadEdge;
23     tie(e1->rot,e2->rot,e3->rot,e4->rot)={e3,e4,e2,e1};
24     tie(e1->nxt,e2->nxt,e3->nxt,e4->nxt)={e1,e2,e4,e3};
25     return e1;
26 }
27
28 void splice(Q a, Q b){
29     swap(a->nxt->rot->nxt,b->nxt->rot->nxt);
30     swap(a->nxt,b->nxt);
31 }
32
33 void del_edge(Q& e, Q ne){
34     splice(e,e->prev()); splice(e->rev(),e->rev()->prev());
35     delete e->rev()->rot; delete e->rev();

```

```

34     delete e->rot; delete e;
35     e=ne;
36 }
37
38 Q conn(Q a, Q b){
39     Q e=edge(a->dest(),b->o,a->rev()->id,b->id);
40     splice(e,a->rev()->prev());
41     splice(e->rev(),b);
42     return e;
43 }
44
45 auto area(pt p, pt q, pt r){return (q-p)%(r-q);}
46
47 // is p in circumference formed by (a,b,c)?
48 bool in_c(pt a, pt b, pt c, pt p){
49     // Warning: this number is 0(max_coord^4).
50     // Consider using int128 or using an alternative method for this
51     // function
52     long double p2=p*p,A=a*a-p2,B=b*b-p2,C=c*c-p2;
53     return area(p,a,b)*C+area(p,b,c)*A+area(p,c,a)*B>EPS;
54 }
55
56 pair<Q,Q> build_tr(vector<pt>& p, int l, int r){
57     if(r-l+1<=3){
58         Q a=edge(p[l],p[l+1],l,l+1),b=edge(p[l+1],p[r],l+1,r);
59         if(r-l+1==2) return {a,a->rev()};
60         splice(a->rev(),b);
61         auto ar=area(p[l],p[l+1],p[r]);
62         Q c=abs(ar)>EPS?conn(b,a):0;
63         if(ar>=-EPS) return {a,b->rev()};
64         return {c->rev(),c};
65     }
66     int m=(l+r)/2;
67     auto [la,ra]=build_tr(p,l,m);
68     auto [lb,rb]=build_tr(p,m+1,r);
69     while(1){
70         if(ra->dest().left(lb->o,ra->o)) ra=ra->rev()->prev();
71         else if(lb->dest().left(lb->o,ra->o)) lb=lb->rev()->next();
72         else break;
73     }
74     Q b=conn(lb->rev(),ra);
75     auto valid=[&](Q e){return b->o.left(e->dest(),b->dest());};
76     if(ra->o==la->o) la=b->rev();

```



```

76 if(lb->o==rb->o) rb=b;
77 while(1){
78     Q L=b->rev()->next();
79     if(valid(L)) while(in_c(b->dest(),b->o,L->dest(),L->next()->dest()))
        del_edge(L,L->next());
80     Q R=b->prev();
81     if(valid(R)) while(in_c(b->dest(),b->o,R->dest(),R->prev()->dest()))
        del_edge(R,R->prev());
82     if(!valid(L)&&!valid(R)) break;
83     if(!valid(L)|| (valid(R)&&in_c(L->dest(),L->o,R->o,R->dest())) b=
        conn(R,b->rev());
84     else b=conn(b->rev(),L->rev());
85 }
86 return {la,rb};
87 }
88
89 vector<vector<int>> del aunay(vector<pt> v){
90     int n=SZ(v); auto tmp=v;
91     vector<int> id(n); iota(ALL(id),0);
92     sort(ALL(id), [&](int l, int r){return v[l]<v[r];});
93     for (int i = 0; i < n; ++i) v[i]=tmp[id[i]];
94     assert(unique(ALL(v))==v.end());
95     vector<vector<int>> g(n);
96     int col=1;
97     for (int i = 2; i < n; ++i) col&=abs(area(v[i],v[i-1],v[i-2]))<=EPS;
98     if(col){
99         for (int i = 1; i < n; ++i) g[id[i-1]].push_back(id[i]),g[id[i]].
            push_back(id[i-1]);
100     }
101     else{
102         Q e=build_tr(v,0,n-1).first;
103         vector<Q> edg={e};
104         for(int i=0;i<SZ(edg);e=edg[i++]){
105             for(Q at=e;!at->used;at=at->next()){
106                 at->used=1;
107                 g[id[at->id]].push_back(id[at->rev()->id]);
108                 edg.push_back(at->rev());
109             }
110         }
111     }
112     return g;
113 }

```

6 Strings

6.1 KMP

```

1 vector<int> kmppre(string& t){ // r[i]: longest border of t[0,i]
2     vector<int> r(t.size()+1);r[0]=-1;
3     int j=-1;
4     for (int i = 0; i < t.size(); ++i){
5         while(j>=0&&t[i]!=t[j])j=r[j];
6         r[i+1]=++j;
7     }
8     return r;
9 }
10 void kmp(string& s, string& t){ // find t in s
11     int j=0;vector<int> b=kmppre(t);
12     for (int i = 0; i < s.size(); ++i){
13         while(j>=0&&s[i]!=t[j])j=b[j];
14         if(++j==t.size())printf("Match at %d\n",i-j+1),j=b[j];
15     }
16 }

```

6.2 Z function

```

1 vector<int> z_function(string& s){
2     int l=0,r=0,n=s.size();
3     vector<int> z(s.size(),0); // z[i] = max k: s[0,k) == s[i,i+k)
4     for (int i = 1; i < n; ++i){
5         if(i<=r)z[i]=min(r-i+1,z[i-1]);
6         while(i+z[i]<n&&s[z[i]]==s[i+z[i]])z[i]++;
7         if(i+z[i]-1>r)l=i,r=i+z[i]-1;
8     }
9     return z;
10 }

```

6.3 Manacher

```

1 int d1[MAXN]; //d1[i] = max odd palindrome centered on i
2 int d2[MAXN]; //d2[i] = max even palindrome centered on i
3 //s aabbaacaabbaa
4 //d1 1111117111111
5 //d2 0103010010301
6 void manacher(string& s){
7     int l=0,r=-1,n=s.size();
8     for (int i = 0; i < n; ++i){

```

```

9     int k=i>r?1:min(d1[l+r-i],r-i);
10    while(i+k<n&& i-k>=0&& s[i+k]==s[i-k])k++;
11    d1[i]=k--;
12    if(i+k>r)l=i-k,r=i+k;
13 }
14 l=0;r=-1;
15 for (int i = 0; i < n; ++i){
16     int k=i>r?0:min(d2[l+r-i+1],r-i+1);k++;
17     while(i+k<=n&& i-k>=0&& s[i+k-1]==s[i-k])k++;
18     d2[i]=--k;
19     if(i+k-1>r)l=i-k,r=i+k-1;
20 }
21 }

```

6.4 Aho-Corasick

```

1 struct vertex {
2     map<char,int> next,go;
3     int p,link;
4     char pch;
5     vector<int> leaf;
6     vertex(int p=-1, char pch=-1):p(p),pch(pch),link(-1){}
7 };
8 vector<vertex> t;
9 void aho_init(){ //do not forget!!
10     t.clear();t.push_back(vertex());
11 }
12 void add_string(string s, int id){
13     int v=0;
14     for(char c:s){
15         if(!t[v].next.count(c)){
16             t[v].next[c]=t.size();
17             t.push_back(vertex(v,c));
18         }
19         v=t[v].next[c];
20     }
21     t[v].leaf.push_back(id);
22 }
23 int go(int v, char c);
24 int get_link(int v){
25     if(t[v].link<0)
26         if(!v||!t[v].p)t[v].link=0;
27     else t[v].link=go(get_link(t[v].p),t[v].pch);

```

```

28     return t[v].link;
29 }
30 int go(int v, char c){
31     if(!t[v].go.count(c))
32         if(t[v].next.count(c))t[v].go[c]=t[v].next[c];
33     else t[v].go[c]=v==0?0:go(get_link(v),c);
34     return t[v].go[c];
35 }

```

6.5 Suffix automaton

```

1 struct state {int len,link;map<char,int> next;}; //clear next!!
2 state st[100005];
3 int sz,last;
4 void sa_init(){
5     last=st[0].len=0;sz=1;
6     st[0].link=-1;
7 }
8 void sa_extend(char c){
9     int k=sz++,p;
10    st[k].len=st[last].len+1;
11    for(p=last;p!=-1&&!st[p].next.count(c);p=st[p].link)st[p].next[c]=k;
12    if(p==st[k].link)st[k].link=0;
13    else {
14        int q=st[p].next[c];
15        if(st[p].len+1==st[q].len)st[k].link=q;
16        else {
17            int w=sz++;
18            st[w].len=st[p].len+1;
19            st[w].next=st[q].next;st[w].link=st[q].link;
20            for(;p!=-1&&st[p].next[c]==q;p=st[p].link)st[p].next[c]=w;
21            st[q].link=st[k].link=w;
22        }
23    }
24    last=k;
25 }
26 // input: abcbcbcb
27 // i,link,len,next
28 // 0 -1 0 (a,1) (b,5) (c,7)
29 // 1 0 1 (b,2)
30 // 2 5 2 (c,3)
31 // 3 7 3 (b,4)
32 // 4 9 4 (c,6)

```

```

33 // 5 0 1 (c,7)
34 // 6 11 5 (b,8)
35 // 7 0 2 (b,9)
36 // 8 9 6 (c,10)
37 // 9 5 3 (c,11)
38 // 10 11 7
39 // 11 7 4 (b,8)

```

6.6 Palindromic Tree

```

1 struct palindromic_tree{
2     static const int SIGMA=26;
3     struct Node{
4         int len, link, to[SIGMA];
5         ll cnt;
6         Node(int len, int link=0, ll cnt=1):len(len),link(link),cnt(cnt)
7         {
8             memset(to,0,sizeof(to));
9         }
10    };
11    vector<Node> ns;
12    int last;
13    palindromic_tree():last(0){ns.push_back(Node(-1));ns.push_back(Node
14        (0));}
15    void add(int i, string &s){
16        int p=last, c=s[i]-'a';
17        while(s[i-ns[p].len-1]!=s[i])p=ns[p].link;
18        if(ns[p].to[c]){
19            last=ns[p].to[c];
20            ns[last].cnt++;
21        }else{
22            int q=ns[p].link;
23            while(s[i-ns[q].len-1]!=s[i])q=ns[q].link;
24            q=max(1,ns[q].to[c]);
25            last=ns[p].to[c]=SZ(ns);
26            ns.push_back(Node(ns[p].len+2,q,1));
27        }
28    };

```

6.7 Suffix array (shorter but slower)

```

1 pair<int, int> sf[MAXN];
2 bool sacomp(int lhs, int rhs) {return sf[lhs]<sf[rhs];}

```

```

3 vector<int> constructSA(string& s){ // O(n log^2(n))
4     int n=s.size(); // (sometimes fast enough)
5     vector<int> sa(n),r(n);
6     for (int i = 0; i < n; ++i)r[i]=s[i];
7     for(int m=1;m<n;m*=2){
8         for (int i = 0; i < n; ++i)sa[i]=i,sf[i]={r[i],i+m<n?r[i+m]:-1};
9         stable_sort(sa.begin(),sa.end(),sacomp);
10        r[sa[0]]=0;
11        for (int i = 1; i < n; ++i)r[sa[i]]=sf[sa[i]]!=sf[sa[i-1]]?i:r[sa[i-1]];
12    }
13    return sa;
14 }

```

6.8 Suffix array

```

1 #define RB(x) (x<n?r[x]:0)
2 void csort(vector<int>& sa, vector<int>& r, int k){
3     int n=sa.size(),top=max(255,n+1);
4     vector<int> f(top,0),t(n);
5     for (int i = 0; i < n; ++i)f[RB(i+k)]++;
6     int sum=0;
7     for (int i = 0; i < top; ++i)f[i]=(sum+=f[i])-f[i];
8     for (int i = 0; i < n; ++i)t[f[RB(sa[i]+k)]]+=sa[i];
9     sa=t;
10 }
11 vector<int> constructSA(string& s){ // O(n log n)
12     int n=s.size(),rank;
13     vector<int> sa(n),r(n),t(n);
14     for (int i = 0; i < n; ++i)sa[i]=i,r[i]=s[i];
15     for(int k=1;k<n;k*=2){
16         csort(sa,r,k);csort(sa,r,0);
17         t[sa[0]]=rank=1;
18         for (int i = 1; i < n; ++i){
19             if(r[sa[i]]!=r[sa[i-1]]||RB(sa[i]+k)!=RB(sa[i-1]+k))rank++;
20             t[sa[i]]=rank;
21         }
22         r=t;
23         if(r[sa[n-1]]==n)break;
24     }
25     return sa;
26 }

```

6.9 LCP (Longest Common Prefix)

```

1 vector<int> computeLCP(string& s, vector<int>& sa){
2     int n=s.size(),L=0;
3     vector<int> lcp(n),plcp(n),phi(n);
4     phi[sa[0]]=-1;
5     for (int i = 1; i < n; ++i)phi[sa[i]]=sa[i-1];
6     for (int i = 0; i < n; ++i){
7         if(phi[i]<0){plcp[i]=0;continue;}
8         while(s[i+L]==s[phi[i]+L])L++;
9         plcp[i]=L;
10        L=max(L-1,0);
11    }
12    for (int i = 0; i < n; ++i)lcp[i]=plcp[sa[i]];
13    return lcp; // lcp[i]=LCP(sa[i-1],sa[i])
14 }

```

6.10 Suffix Tree (Ukkonen's algorithm)

```

1 struct SuffixTree {
2     char s[MAXN];
3     map<int,int> to[MAXN];
4     int len[MAXN]={INF},fpos[MAXN],link[MAXN];
5     int node,pos,sz=1,n=0;
6     int make_node(int p, int l){
7         fpos[sz]=p;len[sz]=l;return sz++;}
8     void go_edge(){
9         while(pos>len[to[node][s[n-pos]]]){
10            node=to[node][s[n-pos]];
11            pos-=len[node];
12        }
13    }
14    void add(int c){
15        s[n++]=c;pos++;
16        int last=0;
17        while(pos>0){
18            go_edge();
19            int edge=s[n-pos];
20            int& v=to[node][edge];
21            int t=s[fpos[v]+pos-1];
22            if(v==0){
23                v=make_node(n-pos,INF);
24                link[last]=node;last=0;

```

```

25    }
26    else if(t==c){link[last]=node;return;}
27    else {
28        int u=make_node(fpos[v],pos-1);
29        to[u][c]=make_node(n-1,INF);
30        to[u][t]=v;
31        fpos[v]+=pos-1;len[v]-=pos-1;
32        v=u;link[last]=u;last=u;
33    }
34    if(node==0)pos--;
35    else node=link[node];
36 }
37 }
38 };

```

6.11 Hashing

```

1 struct Hash {
2     int P=1777771,MOD[2],PI[2];
3     vector<int> h[2],pi[2];
4     Hash(string& s){
5         MOD[0]=999727999;MOD[1]=1070777777;
6         PI[0]=325255434;PI[1]=10018302;
7         for (int k = 0; k < 2; ++k)h[k].resize(s.size()+1),pi[k].resize(s.
8             size()+1);
9         for (int k = 0; k < 2; ++k){
10            h[k][0]=0;pi[k][0]=1;
11            ll p=1;
12            for (int i = 1; i < s.size()+1; ++i){
13                h[k][i]=(h[k][i-1]+p*s[i-1])%MOD[k];
14                pi[k][i]=(1LL*pi[k][i-1]*PI[k])%MOD[k];
15                p=(p*P)%MOD[k];
16            }
17        }
18        ll get(int s, int e){
19            ll h0=(h[0][e]-h[0][s]+MOD[0])%MOD[0];
20            h0=(1LL*h0*pi[0][s])%MOD[0];
21            ll h1=(h[1][e]-h[1][s]+MOD[1])%MOD[1];
22            h1=(1LL*h1*pi[1][s])%MOD[1];
23            return (h0<<32)|h1;
24        }
25    };

```

6.12 Hashing with ll (using __int128)

```

1 #define bint __int128
2 struct Hash {
3     bint MOD=212345678987654321LL,P=17777771,PI=106955741089659571LL;
4     vector<bint> h,pi;
5     Hash(string& s){
6         assert((P*PI)%MOD==1);
7         h.resize(s.size()+1);pi.resize(s.size()+1);
8         h[0]=0;pi[0]=1;
9         bint p=1;
10        for (int i = 1; i < s.size()+1; ++i){
11            h[i]=(h[i-1]+p*s[i-1])%MOD;
12            pi[i]=(pi[i-1]*PI)%MOD;
13            p=(p*P)%MOD;
14        }
15    }
16    ll get(int s, int e){
17        return ((h[e]-h[s]+MOD)%MOD)*pi[s]%MOD;
18    }
19 };

```

7 Flow

7.1 Matching (slower)

```

1 vector<int> g[MAXN]; // [0,n)->[0,m)
2 int n,m;
3 int mat[MAXM];bool vis[MAXN];
4 int match(int x){
5     if(vis[x])return 0;
6     vis[x]=true;
7     for(int y:g[x])if(mat[y]<0||match(mat[y])){mat[y]=x;return 1;}
8     return 0;
9 }
10 vector<pair<int,int> > max_matching(){
11     vector<pair<int,int> > r;
12     memset(mat,-1,sizeof(mat));
13     for (int i = 0; i < n; ++i)memset(vis,false,sizeof(vis)),match(i);
14     for (int i = 0; i < m; ++i)if(mat[i]>=0)r.push_back({mat[i],i});
15     return r;
16 }

```

7.2 Matching (Hopcroft-Karp)

```

1 vector<int> g[MAXN]; // [0,n)->[0,m)
2 int n,m;
3 int mt[MAXN],mt2[MAXN],ds[MAXN];
4 bool bfs(){
5     queue<int> q;
6     memset(ds,-1,sizeof(ds));
7     for (int i = 0; i < n; ++i)if(mt2[i]<0)ds[i]=0,q.push(i);
8     bool r=false;
9     while(!q.empty()){
10        int x=q.front();q.pop();
11        for(int y:g[x]){
12            if(mt[y]>=0&&ds[mt[y]]<0)ds[mt[y]]=ds[x]+1,q.push(mt[y]);
13            else if(mt[y]<0)r=true;
14        }
15    }
16    return r;
17 }
18 bool dfs(int x){
19     for(int y:g[x])if(mt[y]<0||ds[mt[y]]==ds[x]+1&&dfs(mt[y])){
20         mt[y]=x;mt2[x]=y;
21         return true;
22     }
23     ds[x]=1<<30;
24     return false;
25 }
26 int mm(){
27     int r=0;
28     memset(mt,-1,sizeof(mt));memset(mt2,-1,sizeof(mt2));
29     while(bfs()){
30         for (int i = 0; i < n; ++i)if(mt2[i]<0)r+=dfs(i);
31     }
32     return r;
33 }

```

7.3 Hungarian

```

1 typedef long double td; typedef vector<int> vi; typedef vector<td> vd;
2 const td INF=1e100;//for maximum set INF to 0, and negate costs
3 bool zero(td x){return fabs(x)<1e-9;}//change to x==0, for ints/ll
4 struct Hungarian{
5     int n; vector<vd> cs; vi L, R;

```

```

6 Hungarian(int N, int M):n(max(N,M)),cs(n,vd(n)),L(n),R(n){
7     for (int x = 0; x < N; ++x)for (int y = 0; y < M; ++y)cs[x][y]=
        INF;
8 }
9 void set(int x,int y,td c){cs[x][y]=c;}
10 td assign() {
11     int mat = 0; vd ds(n), u(n), v(n); vi dad(n), sn(n);
12     for (int i = 0; i < n; ++i)u[i]=*min_element(ALL(cs[i]));
13     for (int j = 0; j < n; ++j){v[j]=cs[0][j]-u[0];for (int i = 1; i < n
        ; ++i)v[j]=min(v[j],cs[i][j]-u[i]);}
14     L=R=vi(n, -1);
15     for (int i = 0; i < n; ++i)for (int j = 0; j < n; ++j)
16         if(R[j]==-1&&zero(cs[i][j]-u[i]-v[j])){L[i]=j;R[j]=i;mat++;break;}
17     for(;mat<n;mat++){
18         int s=0, j=0, i;
19         while(L[s] != -1)s++;
20         fill(ALL(dad),-1);fill(ALL(sn),0);
21         for (int k = 0; k < n; ++k)ds[k]=cs[s][k]-u[s]-v[k];
22         for(;;){
23             j = -1;
24             for (int k = 0; k < n; ++k)if(!sn[k]&&(j==-1||ds[k]<ds[j]))j
                =k;
25             sn[j] = 1; i = R[j];
26             if(i == -1) break;
27             for (int k = 0; k < n; ++k)if(!sn[k]){
28                 auto new_ds=ds[j]+cs[i][k]-u[i]-v[k];
29                 if(ds[k] > new_ds){ds[k]=new_ds;dad[k]=j;}
30             }
31         }
32         for (int k = 0; k < n; ++k)if(k!=j&&sn[k]){auto w=ds[k]-ds[j];v[
            k]+=w,u[R[k]]-=w;}
33         u[s] += ds[j];
34         while(dad[j]>=0){int d = dad[j];R[j]=R[d];L[R[j]]=j;j=d;}
35         R[j]=s;L[s]=j;
36     }
37     td value=0;for (int i = 0; i < n; ++i)value+=cs[i][L[i]];
38     return value;
39 }
40 };

```

7.4 Dinic

```
1 // Min cut: nodes with dist>=0 vs nodes with dist<0
```

```

2 // Matching MVC: left nodes with dist<0 + right nodes with dist>0
3 struct Dinic{
4     int nodes,src,dst;
5     vector<int> dist,q,work;
6     struct edge {int to,rev;ll f,cap;};
7     vector<vector<edge>> g;
8     Dinic(int x):nodes(x),g(x),dist(x),q(x),work(x){}
9     void add_edge(int s, int t, ll cap){
10         g[s].push_back((edge){t,SZ(g[t]),0,cap});
11         g[t].push_back((edge){s,SZ(g[s])-1,0,0});
12     }
13     bool dinic_bfs(){
14         fill(ALL(dist),-1);dist[src]=0;
15         int qt=0;q[qt++]=src;
16         for(int qh=0;qh<qt;qh++){
17             int u=q[qh];
18             for (int i = 0; i < SZ(g[u]); ++i){
19                 edge &e=g[u][i];int v=g[u][i].to;
20                 if(dist[v]<0&&e.f<e.cap)dist[v]=dist[u]+1,q[qt++]=v;
21             }
22         }
23         return dist[dst]>=0;
24     }
25     ll dinic_dfs(int u, ll f){
26         if(u==dst)return f;
27         for(int &i=work[u];i<SZ(g[u]);i++){
28             edge &e=g[u][i];
29             if(e.cap<=e.f)continue;
30             int v=e.to;
31             if(dist[v]==dist[u]+1){
32                 ll df=dinic_dfs(v,min(f,e.cap-e.f));
33                 if(df>0){e.f+=df;g[v][e.rev].f-=df;return df;}
34             }
35         }
36         return 0;
37     }
38     ll max_flow(int _src, int _dst){
39         src=_src;dst=_dst;
40         ll result=0;
41         while(dinic_bfs()){
42             fill(ALL(work),0);
43             while(ll delta=dinic_dfs(src,INF))result+=delta;
44         }

```

```

45     return result;
46 }
47 };

```

7.5 Min cost max flow

```

1  typedef ll tf;
2  typedef ll tc;
3  const tf INFFLOW=1e9;
4  const tc INFCOST=1e9;
5  struct MCF{
6      int n;
7      vector<tc> prio, pot; vector<tf> curflow; vector<int> prevedge,
          prevnode;
8      priority_queue<pair<tc, int>, vector<pair<tc, int>>, greater<pair<tc,
          int>>> q;
9      struct edge{int to, rev; tf f, cap; tc cost;};
10     vector<vector<edge>> g;
11     MCF(int n):n(n),prio(n),curflow(n),prevedge(n),prevnode(n),pot(n),g(n)
        {}
12     void add_edge(int s, int t, tf cap, tc cost) {
13         g[s].push_back((edge){t,SZ(g[t]),0,cap,cost});
14         g[t].push_back((edge){s,SZ(g[s])-1,0,0,-cost});
15     }
16     pair<tf,tc> get_flow(int s, int t) {
17         tf flow=0; tc flowcost=0;
18         while(1){
19             q.push({0, s});
20             fill(ALL(prio),INFCOST);
21             prio[s]=0; curflow[s]=INFFLOW;
22             while(!q.empty()) {
23                 auto cur=q.top();
24                 tc d=cur.first;
25                 int u=cur.second;
26                 q.pop();
27                 if(d!=prio[u]) continue;
28                 for(int i=0; i<SZ(g[u]); ++i) {
29                     edge &e=g[u][i];
30                     int v=e.to;
31                     if(e.cap<=e.f) continue;
32                     tc nprio=prio[u]+e.cost+pot[u]-pot[v];
33                     if(prio[v]>nprio) {
34                         prio[v]=nprio;

```

```

35         q.push({nprio, v});
36         prevnode[v]=u; prevedge[v]=i;
37         curflow[v]=min(curflow[u], e.cap-e.f);
38     }
39 }
40 }
41 if(prio[t]==INFCOST) break;
42 for (int i = 0; i < n; ++i) pot[i]+=prio[i];
43 tf df=min(curflow[t], INFFLOW-flow);
44 flow+=df;
45 for(int v=t; v!=s; v=prevnode[v]) {
46     edge &e=g[prevnode[v]][prevedge[v]];
47     e.f+=df; g[v][e.rev].f-=df;
48     flowcost+=df*e.cost;
49 }
50 }
51 return {flow,flowcost};
52 }
53 };

```

8 Other

8.1 Mo's algorithm

```

1  int n,sq,nq; // array size, sqrt(array size), #queries
2  struct qu{int l,r,id;};
3  qu qs[MAXN];
4  ll ans[MAXN]; // ans[i] = answer to ith query
5  bool qcomp(const qu &a, const qu &b){
6      if(a.l/sq!=b.l/sq) return a.l<b.l;
7      return (a.l/sq)&1?a.r<b.r:a.r>b.r;
8  }
9  void mos(){
10     for (int i = 0; i < nq; ++i)qs[i].id=i;
11     sq=sqrt(n)+.5;
12     sort(qs,qs+nq,qcomp);
13     int l=0,r=0;
14     init();
15     for (int i = 0; i < nq; ++i){
16         qu q=qs[i];
17         while(l>q.l)add(--l);
18         while(r<q.r)add(r++);
19         while(l<q.l)remove(l++);

```



```

20     while(r>q.r)remove(--r);
21     ans[q.id]=get_ans();
22 }
23 }

```

8.2 Divide and conquer DP optimization

```

1 // O(knlogn). For 2D dps, when the position of optimal choice is non-
  // decreasing as the second variable increases
2 int k,n,f[MAXN],f2[MAXN];
3 void doit(int s, int e, int s0, int e0, int i){
4     // [s,e): range of calculation, [s0,e0): range of optimal choice
5     if(s==e)return;
6     int m=(s+e)/2,r=INF,rp;
7     for (int j = s0; j < min(e0,m); ++j){
8         int r0=something(i,j); // "something" usually depends on f
9         if(r0<r)r=r0,rp=j; // position of optimal choice
10    }
11    f2[m]=r;
12    doit(s,m,s0,rp+1,i);doit(m+1,e,rp,e0,i);
13 }
14 int doall(){
15     init_base_cases();
16     for (int i = 1; i < k; ++i)doit(1,n+1,0,n,i),memcpy(f,f2,sizeof(f));
17     return f[n];
18 }

```

8.3 Dates

```

1 int dateToInt(int y, int m, int d){
2     return 1461*(y+4800+(m-14)/12)/4+367*(m-2-(m-14)/12*12)/12-
3         3*((y+4900+(m-14)/12)/100)/4+d-32075;
4 }
5 void intToDate(int jd, int& y, int& m, int& d){
6     int x,n,i,j;x=jd+68569;
7     n=4*x/146097;x-=(146097*n+3)/4;
8     i=(4000*(x+1))/1461001;x-=1461*i/4-31;
9     j=80*x/2447;d=x-2447*j/80;
10    x=j/11;m=j+2-12*x;y=100*(n-49)+i+x;
11 }
12 int DayOfWeek(int d, int m, int y){ //starting on Sunday
13     static int ttt[]={0, 3, 2, 5, 0, 3, 5, 1, 4, 6, 2, 4};
14     y-=m<3;
15     return (y+y/4-y/100+y/400+ttt[m-1]+d)%7;

```

```

16 }

```

8.4 C++ stuff

```

1 // double inf
2 const double DINF=numeric_limits<double>::infinity();
3 // Custom comparator for set/map
4 struct comp {
5     bool operator()(const double& a, const double& b) const {
6         return a+EPS<b;}
7 };
8 set<double,comp> w; // or map<double,int,comp>
9 // Iterate over non empty subsets of bitmask
10 for(int s=m;s;s=(s-1)&m) // Decreasing order
11 for (int s=0;s=s-m&m;) // Increasing order
12 // Return the numbers the numbers of 1-bit in x
13 int __builtin_popcount (unsigned int x)
14 // Returns the number of trailing 0-bits in x. x=0 is undefined.
15 int __builtin_ctz (unsigned int x)
16 // Returns the number of leading 0-bits in x. x=0 is undefined.
17 int __builtin_clz (unsigned int x)
18 // x of type long long just add 'll' at the end of the function.
19 int __builtin_popcountll (unsigned long long x)
20 // Get the value of the least significant bit that is one.
21 v=(x&(-x))

```

8.5 Interactive problem tester template

```

1 #Easier method with bash commands:
2 #mkfifo fifo
3 #(. /solution < fifo) | (. /interactor > fifo)
4
5 # tester for cf 101021A (guess a number, queries: is it >=k?)
6 import random
7 import subprocess as sp
8 seed = random.randint(0, sys.maxint);random.seed(seed)
9 n=random.randint(1,1000000)
10 try:
11     p=sp.Popen(['./a.out'],stdin=sp.PIPE,stdout=sp.PIPE)
12     it=0
13     s=p.stdout.readline()
14     while it<25 and s and s[0]!='!':
15         k=int(s)
16         assert k>=1 and k<=1000000

```

```

17     if n>=k: p.stdin.write('>=\\n')
18     else: p.stdin.write('<\\n')
19     s=p.stdout.readline()
20     it+=1
21     assert s and s[0]=='!'
22     k=int(s.split()[1])
23     assert k==n
24 except:
25     print 'failed_with_seed_%s' % seed
26     raise

```

8.6 Max number of divisors up to 10^n

```

1 (0,1) (1,4) (2,12) (3,32) (4,64) (5,128) (6,240) (7,448) (8,768)
  (9,1344) (10,2304) (11,4032) (12,6720) (13,10752) (14,17280)
  (15,26880) (16,41472) (17,64512) (18,103680)

```

8.7 Template

```

1 #include <bits/stdc++.h>
2 #ifdef DEMETRIO
3 #define deb(...) fprintf(stderr,__VA_ARGS__)
4 #define deb1(x) cerr << #x << " = " << x << endl
5 #else
6 #define deb(...) 0
7 #define deb1(x) 0
8 #endif
9
10 // --- DEBUGGING TOOLS ---
11 void void_debug() { std::cerr << std::endl; }
12 template<typename Head, typename... Tail>
13 void void_debug(const Head& H, const Tail&... T) { std::cerr << "┐" << H
  ; void_debug(T...); }
14
15 #ifdef LOCAL
16 #define debug(...) std::cerr << "[" << #__VA_ARGS__ << "]:", void_debug(
  __VA_ARGS__)
17 #else
18 #define debug(...) 42
19 #endif
20
21 /*
22 Uso en competencia:

```

```

23 Compila localmente con el flag -DLOCAL para activar las salidas por cerr
  , por
24 ejemplo:
25     g++ -std=c++17 -O2 -Wall -Wextra -DLOCAL main.cpp -o main
26
27 Dentro del codigo puedes escribir:
28     debug(variable_x, mi_vector[0], "llegue aqui");
29
30 Cuando LOCAL no esta definido, como ocurre normalmente al enviar al juez
  en
31 linea, debug(...) se expande a una expresion inocua y no imprime nada.
  Esto
32 mantiene las trazas fuera del output oficial sin tener que borrar
  llamadas de
33 debug antes de enviar gracias a #ifdef LOCAL. Si tu entorno usa
34 #ifndef ONLINE_JUDGE, puedes mantener la misma idea: compila localmente
  con
35 -DLOCAL y envia sin ese flag.
36 */
37
38 #define SZ(x) ((int)x.size())
39 using namespace std;
40 typedef long long ll;
41
42 int main(){
43     return 0;
44 }

```